

COURTESY OF SILAS OTIENO
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BAC 200: ACCOUNTING FOR ASSETS

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CHAPTER ONE

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CONCEPTUAL FRAMEWORK FOR FINANCIAL ACCOUNTING AND REPORTING

Objectives

By the end of this lesson, you should be able to: Understand the objectives of financial reporting Describe the usefulness of a conceptual framework Identify the qualitative characteristics of accounting information Describe the basic assumptions of accounting Describe the impact that constraints have on reporting accounting information.

Introduction

A conceptual framework is like a constitution. It is a coherent system of interrelated objectives and fundamentals that can lead to consistent standards and that prescribe the nature, function and limits of financial accounting in financial statements. A conceptual framework is essential in that to be useful, standard setting should be built on and relate to an established body of concepts and objectives. In addition new and emerging practical problems should be quickly solved by reference to an existing framework of basic theory.

Development of a conceptual framework

Over the years numerous organisations, committees and interested individuals have developed and published their own conceptual frameworks. But no single framework has been universally accepted and relied on in practice. In general however a conceptual framework for financial reporting is based on the three levels.

- The first level which relates to the goals and purposes of accounting thus basically dealing with the objectives of financial reporting.
- The second level that deals with the qualitative characteristics of accounting information and the elements of financial statements. This level forms the bridge between the first level and the third.
- The third level that deals with the recognition and measurement aspects of accounting information.

First level:

Objectives of financial reporting and financial statements

The objectives of financial reporting and financial statements are derived from the needs of the external users of accounting information. Financial statements intended to serve all external users often are called general-purpose financial statements. Stating the objectives of financial statements would be simpler if all external users had the same needs and interests, but they do not. For example, a banker considering the granting of a 91-day loan is primarily interested in the short-run debt-paying ability of the business enterprise, whereas the long-term investor in common stock is

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more concerned with earning capacity, potential growth in earnings per share, and the ability of the enterprise to survive as a going concern.

Because general-purpose financial statements serve a variety of users, the needs of some users receive more emphasis than the needs of others. In present-day accounting practice the needs of the potential investor or creditor are subordinated to those who have already invested resources in the enterprise. This emphasis lead management of the enterprise to stress the uses made of the resources entrusted to it. Higher standards of measurement and reporting, along with a significant expansion of the amount of information disclosed, have been foremost among the new needs of users of financial statements.

In general financial reporting should provide information:

- That is useful to present and potential investors and creditors and other users in making rational investment, credit, and similar decisions. The information should be comprehensible to those who have a reasonable understanding of business and economic activities and are willing to study the information with reable diligence.
- To help present and potential investors and creditors and other users in assessing the amounts, timing, and uncertainty of prospective cash receipts from dividends or interest and the proceeds from the sale, redemption, or maturity of securities or loans.
- About the economic resources of an enterprise, the claims to those resources, and the effects of transactions, events, and circumstances that change resources and claims to those resources.
- About an enterprise's financial performance during a period. Investors and creditors often use information about the past to help in assessing the prospects of an enterprise.
- About how an enterprise obtains and spends cash, about its borrowing and repayment of borrowing, about its capital transactions, including cash dividends and other distributions of enterprise resources to owners, and about other factors that may affect an enterprise's liquidity or solvency.
- About how management of an enterprise has discharged its stewardship responsibility to owners (stockholders) for the use of enterprise resources entrusted to it.

In summary, the primary focus of financial reporting is information about an enterprise's performance provided by measures of earnings and its components.

Second level

Qualitative Characteristics of Accounting Information

The qualitative characteristics underlying good accounting information can be divided into:

- Primary characteristics

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- Secondary characteristics

Primary characteristics

Accounting information should possess two primary qualitative characteristics:

- Relevance
- Reliability **Relevance**

To be relevant accounting information must be capable of making a difference in a decision. Relevant information helps users make predictions about the ultimate outcome of past, present and future events, that is it has predictive value. For example, information concerning past dividends declared by a corporation enables investors to predict the prospects of dividends in future years. Relevant information also helps users confirm or correct prior expectations that is, it has feedback value. For example, information on net income for the first three-quarters of a fiscal year enables investors to evaluate a prior estimate of net income for the entire fiscal year. Accounting information generally is not relevant unless it is timely, that is, unless it is available to decisions makers before it becomes too dated to influence the decision thus timeliness is a primary ingredient.

Reliability

Accounting information is reliable if it is reasonably free from error and bias and faithfully represents what it purports to present. To be reliable, information must be verifiable and must possess representational faithfulness, or validity. Reliability is a necessity for individuals who have neither the time nor expertise to evaluate factual contents of the information.

Neutrality

Neutrality is defined as the absence in reported information of bias intended to attain a predetermined result or to induce a particular mode of behavior. Because of the many users of general-purpose financial statements, freedom from bias is essential. For example, financial statements designed solely to influence the actions of investors could be damaging to the needs of creditors, another major user group for financial statements. If accounting information were free from bias, different accountants working independently and following the same measurement techniques would reach the same conclusions.

Representational faithfulness

This means that the numbers and descriptions represent what really existed or happened. For instance if a company reports sales amounting to Kshs 200M in its income statement when actually the sales amounted to Kshs 170M, the company's statement are then not a faithful representation of the real facts.

Verifiability

To verify means to prove something to be true by examination of evidence of underlying facts. In most cases, actual costs provide the most reliable data capable of being independently verified. Supporting documents showing the details of completed "arm's-length" transactions provide clear evidence that may be verified. Verifiability is clearly demonstrated when independent parties using the same measurement methods obtain similar results.

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However, financial statements are not completely factual; estimates on such matters as the economic life of plant assets, the net realizable value of inventories, and the collectibility of accounts receivable are inherent in the accounting process. The reliability quality calls for accountants to adhere as closely as possible to objectively verifiable evidence.

Secondary characteristics

The secondary qualitative characteristics of accounting information include:

- Comparability
- consistency

Comparability

Information that has been measured and reported in a similar manner for different enterprises is considered comparable. Comparability of the financial statements of a business enterprise from one accounting period to the next is essential of favourable and unfavourable trends in the enterprise are to be identified. If the financial statements for the current accounting period show larger earnings than for the preceding period, the user assumes that operations have been more profitable. However, if a material change in an accounting principle has occurred, the reported increase in any improvement in the underlying business activity. Comparability among financial statements of business enterprises in the same industry also is a useful quality. However, differences in the operating policies among such enterprises result in the adoption of various accounting practices in such areas as valuation of inventories and depreciation of plant assets. Thus, comparability of financial statements of enterprises in the same industry is difficult to achieve

Consistency

When an entity applies the same accounting treatment to similar events from period to period the entity is considered to be consistent in its use of accounting standards. Consistent application of accounting principles for a business enterprise from one accounting period to the next is needed in order that the financial statements of successive periods will be comparable. However, the consistency principle does not mean that a particular method of accounting, once adopted, should not be changed. Accounting principles and methods change in response to changes in the environment of accounting. When an accounting change is desirable, it should be made, together with disclosure of the change and its effect in money measurement on the reported net income of the accounting period in which the change is made.

Constraints to useful accounting information

In providing information with the qualitative characteristics that make it useful, two overriding constraints must be considered;

- Materiality
- Cost benefit considerations

However two other less dominant yet important constraints need also to be reflected;

- Industry practices
- conservatism

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Materiality

Disclosure is necessary in financial statements or in notes to the financial statements only for material matters. The meaning of materiality in an accounting context is a state of relative importance. Items that are trifling in amount need not be treated in strict accordance with accounting theory but rather should be handled in the most economical manner. In theory, the cost of a new pencil sharpener should be capitalized and depreciated over its economic life. As a practical matter, the expense of making such allocations of cost would exceed the cost of the pencil sharpener and would represent an unjustifiably wasteful accounting policy. In deciding on the materiality of an item in terms of financial statement disclosure, accountants should consider whether knowledge of the item would be likely to influence the decisions of users of financial statements. That which is material for one business enterprise may not be for another. For a small enterprise, an uninsured loss, say Kshs 50,000, might be considered as material; for a large enterprise it would not be material.

Cost-Benefit Considerations.

Before a decision is made to develop a standard, the standard setting board needs to satisfy itself that the matter to be ruled on represents a significant problem and that a standard that is promulgated will not impose costs on the many for the benefit of a few. If the proposal passes that first test, a second test may subsequently be useful. There are usually alternative ways of handling an issue. Is one of them less costly and only slightly less effective? Even if absolute magnitudes cannot be attached to costs and benefits, a comparison between alternatives may yet be possible and useful.

Conservatism

Although not a qualitative characteristic of accounting information, conservatism is a concept that may be discussed in connection with reliability. Many accounting measurements do not have a single “correct answer”; a choice must be made among alternative assumptions under conditions of uncertainty. The concept of conservatism holds that when reasonable support exists for alternative accounting methods and for different measurement techniques, accountants should select the method or technique with the least favourable effect on net income and financial position in the current accounting period.

Note

The concept of conservatism should not be distorted to the point of deliberate understatement however; the judicious use of conservatism in accounting may help to prevent the catastrophes that have befallen many investors and employees with excessively optimistic accounting policies.

Industry practices

The peculiar nature of some business concerns sometimes requires a departure from the stated accounting theory. For instance, agricultural crops are often reported at market values because it is costly to develop accurate cost figures on individual crops. Whenever a variation emerges, it

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should be determined whether some peculiar feature of the type of business involved can explain it before criticising the procedure followed.

Elements of financial statements

An important aspect of developing any theoretical structure is the body of basic elements or definitions to be included in the structure. At present, accounting uses many terms that have peculiar and specific terms that have peculiar and specific meanings. These terms constitute the language of business or the accounting jargon. The most commonly used terms include:

- Assets - these are probable future economic benefits obtained or controlled by a particular entity as a result of past transactions or events.
- Liabilities -these are probable future sacrifices of economic benefits arising from present obligations of a particular entity to transfer assets or provide services to other entities in the future as a result of past transactions or events.
- Equity- this is the residual interest in the assets of an entity that remains after deducting its liabilities.
- Revenues- these are inflows or other enhancements of assets of an entity or settlement of its liabilities during a period from delivering or producing goods, rendering services, or other activities that constitute the entity's ongoing major or central operations.
- Expenses- these are outflows or other using up of assets or incurrences of liabilities during a period from delivering or producing goods, rendering services, or other activities that constitute the entity's ongoing major or central operations.
- Gains- these are increases in equity (net assets) from peripheral or incidental transactions of an entity and from all other transactions and other events and circumstances affecting the entity during a period except those that result from revenues or investments by owners.
- Losses- these are decreases in equity (net assets) from peripheral or incidental transactions of an entity and from all other transactions and other events and circumstances affecting the entity during a period except those that result from expenses or distributions to owners.

Third level

Recognition and measurement concepts

The third level of the framework consists of concepts that implement the basic objectives of financial reporting. These concepts explain which, when, and how financial elements and events should be recognised, measured and reported by the accounting system. The concepts serve as guidelines in developing rational responses to controversial financial reporting issues. They are classified as assumptions and principles.

Basic assumptions

The term generally accepted accounting principles has long been used in financial accounting. This term also is used by Certified Public Accountants in their audit reports to indicate whether

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the business enterprise being audited has prepared its financial statements in an acceptable manner, so that they may be compared with the prior year's statements and to some extent with the statements of other enterprises. The principles of accounting are not rooted in the laws of nature, as are the physical sciences. Therefore, accounting principles must be developed in relation to the stated objective of financial reporting and financial statements.

Four main assumptions underlie the financial accounting structure. They include;

Economic entity assumption

This assumption implies that the activities of a business enterprise should be kept separate and distinct from its owners and any other business unit. The entity concerned does not necessarily refer to a legal entity. A parent company and its subsidiary are separate legal entities but merging their activities for accounting and reporting purposes does not violate the economic entity assumption.

Going concern assumption

Although accountants do not believe that business firms will last indefinitely, they do expect them to last long enough to fulfil their objectives and commitments. The implications of this assumption are profound. The assumption of continued existence provides the logical basis for recording probable future economic benefits as assets and probable future outlays as liabilities. The continuity principle implies not permanence of existence but simply that a business enterprise will continue in existence long enough to carry out present plans and meet contractual commitments. This principle affects the classification of assets and liabilities in a balance sheet. Because it is assumed that assets will be used and obligations paid in the normal course of operation, no attempt is made to classify assets and liabilities in terms of their ultimate disposition or legal priority in case of liquidation.

Monetary unit assumption

The monetary unit assumption means that accountants assume money to be a useful standard measuring unit for reporting the effects of business transactions. Money is used as the common denominator throughout the accounting process. Some of the information necessary to give a comprehensive picture of a business enterprise is difficult or impossible to quantify and express in money or other units of measurement. Examples are the competence, health, and morale of management and employees, and the effect of the operations of the enterprise on the natural environment. However, if information is to be included in financial statements, it must be expressed in monetary terms. If such measurement is not practicable, a possible alternative method of communication is to use notes to the financial statements.

Periodicity assumption

This assumption is also referred to as the time period assumption. It implies that the economic activities of an enterprise can be divided into artificial time periods. These time periods vary, but the most common are monthly, quarterly and yearly. The most accurate way to measure the results of enterprise activity would be to measure them at the time of the enterprise's eventual liquidation. Businesses, investors, government and other user groups, however cannot wait that long for such

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information. Users need to be appraised of performance and economic status on a timely basis so that they can evaluate and compare firms. Therefore information must be reported periodically.

Basic accounting principles

Four basic principles of accounting are used to record transactions. They include;

Matching Principle

The matching principle means that after the revenue (accomplishment) for an accounting period has been determined, the costs (effort) associated with this revenue must be deducted from revenue to measure net income. The term matching refers to the close relationship that exists between certain costs and the revenue recognized as a result of incurring those costs. The matching of a business enterprise's expenses (or expired costs) with its revenue for an accounting period is the primary activity in the measurement of the results of the enterprise's operations for that period. For example, expenditures for advertising attract customers and generate sales. The outlay for the advertising is one of the expenses to be deducted from the revenue of an accounting period. Similarly, the recognition of doubtful accounts' expense illustrates the importance of the accounting period in the matching of expenses and revenue. Doubtful accounts' expense is caused by selling goods or services on credit to customers who fail to pay their bills. To match this expense with the related revenue, the expense must be recorded and deducted from revenue in the accounting period in which the sales are made and recorded, even though the receivables are not determined to be uncollectible until the following period. The use of estimates is necessary in this and in many other situations in order to implement the matching principle.

Disclosure Principle

The disclosure principle requires that financial statements be complete in the sense of including all information necessary to users of the statements. If the omission of certain information would cause the financial statements to be misleading, disclosure of such information is essential. Published financial statements include detailed notes that are considered to be an integral part of the statements. However, disclosures in the notes should supplement the information in the body of the financial statements and should not be used to correct improper presentation of information in the body of the statements. Typical examples of information often disclosed in notes to financial statements include the following: a summary of significant accounting policies, related party transactions, descriptions of stock option and pension plans, status of litigation in which the business enterprise is a party, amount and nature of loss contingencies and commitments, and terms and status of proposed business combinations.

Note:

The concept of disclosure applies not only to transactions and events that have occurred during the accounting period covered by the financial statements, but also to material subsequent events that occur after the balance sheet date but before the financial statements are released. Such events are disclosed in a note to the financial statements.
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Historical cost principle

Under this principle, assets are initially recorded in the accounts at cost, and no adjustment is made to this valuation in later periods, except to allocate a portion of the original cost to expense as the assets expire. At the time an asset is originally acquired, cost represents the fair market value of the goods or services exchanged as evidenced by an arm's length transaction. With the passage of time, however, the fair market value of such assets as buildings may change greatly from the historical cost. These later changes in fair market value in the balance sheet at historical cost.

Note

Despite the criticism levelled by an increasing number of accounting professionals, this principle has continued to be applied on the basis of its objectivity. It is argued that it is important that users of accounting information have confidence in financial statements, and this confidence is best maintained if accountants recognise changes in assets and liabilities only on the basis of completed transactions. Objective evidence generally exists to support cost, but evidence supporting current values may be less readily available.

Revenue recognition principle

This principle states that revenue should be recognised when:

- Realised, or realisable
- Earned.

Revenues are realised when goods or services or other assets are exchanged for cash or claims to cash. Revenues are realisable when assets received or held are readily convertible into cash or claims to cash. Assets are readily convertible when they are saleable or interchangeable in an active market at readily determinable prices without significant additional cost.

In addition to the first test, revenues are not recognised until earned. They are considered earned when the entity has substantially accomplished what it must do to be entitled to the benefits represented by the revenues. Generally, an objective test; confirmation by a sale to independent interests is used to indicate the point at which revenue is recognised. Usually only at the date of sale is there an objective and verifiable measure of revenue, that is, the sales price.

Exercises

1. Are generally accepted accounting principles equally applicable to the fields of financial accounting and managerial accounting? Explain
2. Define general purpose financial statements and point out the limitations of such statements
3. Briefly describe the objectives of financial reporting.
4. The two primary qualities of accounting information are relevance and reliability. Are these qualities as likely to be present in a forecast of future earnings as in an income statement? Explain.
5. What is meant by the going concern principle of accounting? How does it affect the valuation of assets? When is this principle not applicable?

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Courtesy of Rodgers Wesonga

CHAPTER TWO

CASH

Objectives

By the end of this lesson, you should be able to: Identify items considered cash
Indicate how cash and related items are reported To fully understand the procedure of preparing a bank reconciliation statement.

Introduction

Cash can be defined as a medium of exchange that a bank will accept for deposit and immediately credit to the depositor's account. It includes; currency notes and coins, personal cheques, bank drafts, money orders, credit cards as well as money in deposit with banks. Deposits with a trustee for instance a bond sinking fund that is not under the control of management the business enterprise should not be included as part of cash. Certificates of deposit are generally classified as short-term investment rather than as cash because they are not available for immediate withdrawal. Strictly speaking, savings deposits may also not be withdrawn without prior notice to the bank but banks rarely enforce this requirement. Consequently savings deposits usually are viewed as cash.

In summary the criteria generally used to define cash is that the item as a medium of exchange must be available immediately for payment of current debts and be free from any contractual restrictions that would prevent management of the business enterprise from using them to pay its creditors.

Management of cash

Management of cash is of major importance in any business enterprise because cash is a means of acquiring goods and services. In addition careful scrutiny of cash transactions is required because cash may easily be misappropriated. The responsibilities of management with respect to cash include:

- To ensure that there is sufficient cash to carry on the operations of the enterprise
- To invest idle cash
- To prevent loss of cash due to theft or misappropriation

Cash forecasting is necessary for the proper planning of future operations and to assure that cash is available when needed but cash in hand is not excessive. Internal controls are necessary to assure that cash is used for proper business practices and not wasted, misused or stolen. **Internal controls** The purpose of a system of internal controls is to assure that assets that belong to the business enterprise are;

- Received when tendered
- Protected when in the custody of the business
- Used only for authorised business practices.

Such a system consists of administrative and accounting controls.

Accounting controls

These controls are designed to provide reasonable assurance that;

- Transactions are executed in accordance with the management's general or specific authorisation

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- Transactions are recorded as necessary to permit preparations of financial statements in conformity with generally accepted accounting principles or any other criteria applicable to such statements
- Access to assets is permitted only in accordance with management authorisation.

A system of internal controls is not designed primarily to detect errors but rather to reduce the possibility of errors or dishonesty occurring.

System of internal check

This is an effective system of control where no single person should carry out all phases of a business transaction from beginning to the end. For instance, if one person were permitted to order merchandise, receive it, write a cheque in payment and record the transaction in the accounting records there would be no protection either against frauds or errors. In large entities separate and independent departments are established for such functions such as purchasing, receipt of merchandise, selling, accounting and finance so as to ensure that no one department controls all phases of a transaction.

Controlling cash receipts and payments.

The objective sought in the control of cash receipts is to ensure that all cash that is receivable by the business is collected and recorded without loss. The system of controlling cash payment should be designed to ensure that no unauthorised payments are made. Keeping it in a safe, depositing in the bank and through the use of special imprest cash funds safeguards cash.

Reconciliation of bank balances

The balance indicated in a bank statement rarely agrees with the bank balance as per the trader's cashbook. A bank reconciliation statement is a detailed statement reconciling at a given date the cash balance reported by the bank with that shown in the records of the business. Its purpose is thus to explain the difference that exists between the two figures. Reasons that may lead to this discrepancy include:

- Deposits in transit- end of month deposits of cash recorded on the depositor's books in one month are received and recorded by the bank in the following month.
- Outstanding cheques- cheques written by the depositor are recorded when written but have not yet been presented to the bank for payment until after a later date thus have not been recorded by the bank by the time the statement is prepared.
- Dishonoured cheques
- Bank charges- charges recorded by the bank against the depositor's balance for such items as cheque processing, ledger fees or safe deposit rental. The depositor is normally not aware of these charges until the receipt of the bank statement.
- Bank credits - collections, payments or deposits by the bank for the benefit of the depositor that may be unknown to the depositor until receipt of the bank statement,
-
- for instance interest earned on certificates of deposit and direct transfers.

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- Standing orders- orders by the account holder to the bank authorising the bank to make regular payments of fixed amounts at stated date to certain people.
- Bank or depositor's errors- errors on the part of the bank or depositor cause the bank balance to disagree with the depositor's bank balance as per the cashbook.

Illustration

Mr. Ojwang, a sole trader received his bank statement for the month of June 2001. At that date the statement showed a credit balance of Kshs 49,320 whereas his cashbook showed a debit balance of Kshs 36,080. After a lengthy investigation into the cause of the difference the following was discovered;

- Cheques totalling Kshs 61,450 issued by Ojwang did not appear in the bank statement
- The bank credited cheques totalling Kshs 39,450 deposited on the last day of the month on 5th July.
- Bank charges for the month amounted to Kshs 2,410
- The bank returned a cheque for Kshs 6,050 deposited by Ojwang to the bank on 23rd June unpaid on 4th July.
- The bank made payments of Kshs 8000 to Ojwang's landlord and another of Kshs 2000 to an insurance company as part of Ojwang's standing orders with bank
- A debtor remitted Kshs 9,700 directly to Ojwang's account.

Required

Prepare a statement reconciling the cashbook balance and the bank statement balance as at 30th June 2001

Solution

Ojwang's bank reconciliation statement As at 30/06/2001

	Kshs	Kshs
Balance as per the cashbook	Add	36,080
Outstanding cheques	61,450	
Direct remittances	<u>9,700</u>	
		<u>71,150</u>
		107,230
Deduct		
Uncredited cheques	39,450	
Standing orders (8000+2000)	10,000	
Dishonoured cheque	6,050	

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Bank charges	<u>2,410</u>
	<u>57,910</u>
Balance as per the bank statement	49,320

Note:

When the balance as per the bank statement is reconciled towards the cashbook balance, The procedure reverses with additions becoming the deductions and the deductions the additions .

Activity

Prepare the bank reconciliation statement starting with the balance as per the bank statement

Sometimes, it may be necessary to adjust the cashbook before preparing the reconciliation statement. To carry out this task, all the items appearing in the bank statement that need also to feature in the trader's cashbook and have not been recorded need to be transferred. The adjusted cashbook balance is then reconciled towards the bank statement balance.

Illustration

Using the previous example on Mr. Ojwang adjust his cashbook and consequently prepare the bank reconciliation statement.

Solution

	Kshs	Kshs
Balance as per the cashbook before adjustment		36,080
Add (Dr.)		
Direct remittance		<u>9,700</u>
		45,780
Less (Cr.)		
Bank charges	2,410	
Dishonoured cheque	6,050	
Standing orders	<u>10,000</u>	
		<u>18,460</u>
Adjusted cashbook balance		<u>27,320</u>

Ojwang's bank reconciliation statement

As at 30/06/2001

	Kshs	Kshs
Adjusted cashbook balance	27,320	
Add:		
Outstanding cheques		<u>61,450</u>

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88,770

Less:

Uncredited cheques	<u>39,450</u>
Balance as per the bank statement	<u>49,320</u>

Exercises

1. What are the normal contents of cash?
2. What is a system of internal control?
3. Differentiate between internal administrative controls and internal accounting controls.
4. What are the functions of the bank reconciliation statement?
5. How should a material cash overdraft be reported in the balance sheet?
6. You have recently been employed in a medium size company and deployed in the accounts department. Your head of section has given you the following extract from the cashbook for the month of April 2010:

	Sh.		Sh.
Receipts during the month	293,800	Balance brought forward (1.4.2010)	152,200
Balance carried forward (30.4.2010)	<u>110,800</u>	Payments during the month	<u>252,400</u>
	<u>404,600</u>		<u>404,600</u>

The head of section further informs you that all receipts are banked intact and all payments are made by cheque. On investigation, you discover the following:

1. Bank charges and commissions amounting to Sh. 27,200 entered on the bank statement had not been entered in the cashbook.
2. Cheques drawn amounting to Sh. 53,400 had not been presented to the bank for payment.
3. Cheques received totaling Sh. 152,400 had been entered in the cashbook and paid into the bank, but had not been credited by the bank until May 2010.
4. A cheque for Sh. 4,400 had been entered as a receipt in the cashbook instead of a payment.
5. A cheque for Sh. 5,000 had been debited by the bank by mistake.
6. A cheque received for Sh. 16,000 had been returned unpaid. No adjustment had been made in the cashbook.
7. All dividends receivable are credited direct to the bank account. During the month of April 2010 Dividends totaling Sh. 12,400 were credited by the bank and no entries had been made in the cashbook.
8. A cheque drawn for Sh. 1,200 had been incorrectly entered in the cash book as Sh. 13,200.
9. The balance brought forward should have been Sh. 142,200.
10. The bank statement as at 30 April 2010 showed an overdraft of Sh. 232,400.

Required:

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- (i) The adjusted cashbook as at 30 April 2010.
- (ii) Bank reconciliation statement as at 30 April 2010.

7. The following data pertaining to the cash transactions and bank account of Kim Company for September, Year 4, are available to you:

- a) Cash balance per accounting records , Sep 30 Year 4
Sh67,500
- b) Cash balance per accounting bank statement, Sep 30 Year 4
308,383
- c) Bank service charge for September
1,100
- d) Debit memo for printed cheques delivered by the bank
300
- e) Deposit of Sep 30 not recorded by bank until Oct 1
48,700
- f) Outstanding cheques, Sep 30, Year 4
81,283
- g) Proceeds of a bank loan on Sep 30 not recorded in the accounting records
(interest payable on maturity)
300,000
- h) Proceeds from Note Receivable , principal amount sh8,000
8,100
- i) Cheque no 1012 to a creditor entered in the accounting records as sh 18,791;
Deducted in the bank statement in the correct amount
17,891
- j) Stolen cheque lacking an authorized signature deducted from Kim's account in
Error
86,700
- k) Cheque from a debtor returned by the bank marked NSF 12,600

Required:

- i) Prepare a bank reconciliation as at September 30, Year 4 ii)
Prepare journal entry(ies) to adjust the accounting records

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Courtesy of Rodgers Wesonga

CHAPTER THREE: SHORT TERM INVESTMENTS

Introduction

To achieve efficient use of all resources, management of a business enterprise frequently turns unproductive cash balance into productive resources through the purchase of short-term investments. Such investments held by an enterprise for the purpose of earning a return on cash resources are characterised by their saleability at a readily determinable price. Stocks and bonds, which are not widely held or frequently traded usually, do not meet the marketability test; consequently, securities of this type are not considered under short-term investments. Investments in securities of other companies purchased as a means of exercising influence or control over the operations of such companies should not be considered as part of short-term investments.

In summary, short-term investments, which are classified in the balance sheet, as current assets must be readily saleable and should not be, held for purpose of bolstering business relations with the issuing entity. On the other hand, there is no requirement that the investment be held for a limited time only or that management express its intent as to the duration of the holding.

The objectives of acquiring short-time investments are twofold:

- (i) to maximise the return on invested capital
- (ii) to minimise the risk of loss from price fluctuations.

Recording transactions in short-term investments

At acquisition, short-term investments are recorded at cost, the price of the item being in the market plus any costs incident to the acquisition, such as brokerage commissions and transfer taxes. Bonds acquired between interest date are traded on the basis of the market price plus the interest accrued

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since the most recent interest payment. The accrued interest is a separate asset, which is purchased with the bonds.

Note:

The cost of these two assets should be separated to achieve a clear picture of the results of the investment in bonds.

Illustration

On March 31 2002, Ruiru Limited placed an order with a broker to buy 100,000, 10% Safaricom bonds which mature on January 31, 2005, with interest dates July 31 and January 31. The bonds were purchased on the same day at 104 plus accrued interest of Kshs 166,667 for two months. The brokerage commission was Kshs 1,000,000. Compute the total cost of the bonds and the total cash outlay.

Solution

	Kshs
Market price of bonds (Kshs 104,000*100)	10,400,000
Add: Brokerage commission	<u>1,000,000</u>
Total cost of bonds	<u>11,400,000</u>
Add: accrued interest for 2 months on Kshs 10M at 10%	<u>166,667</u>
Total cash outlay	<u>11,566,667</u>

Journal entries required in the books of Ruiru Limited

	Kshs	Kshs
Short-term investments	11,400,000	
Accrued interest receivable	166,667	
Cash		11,566,667

Computation of cash received from sale of Bonds

Assume that on 30th June 2001, Ruiru Limited sold the Safaricom bonds at 105 $\frac{3}{4}$ plus accrued interest for five months. Compute the cash received from sale of the bonds, after brokerage commission of Kshs 1M.

Solution

	Kshs
Market price of bonds (105,000*100)	10,575,000
Less: Brokerage commission	<u>(1,000,000)</u>
Proceeds on sale of bond	9,575,000
Add: accrued interest for 5 months at 10%	<u>416,667</u>
Total cash received	<u>9,991,667</u>

Journal entries required in the books of Ruiru Limited

	Kshs	Kshs
Cash	9,991,667	
Short term investment		11,400,000
Accrued interest receivable		166,667
Interest revenue		250,000

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Loss on sale of short-term Investment 1,825,000

Balance sheet presentation of cash and short term investments

Cash is the most liquid asset that a firm owns, in the sense that it is most easily converted to other assets and services. This characteristic justifies its position as the first item in the current assets section of the balance sheet. Short-term investments rank next to cash in liquidity and thus are listed below cash in the current asset section of the balance sheet.

Note:

Whether short-term investments are recorded at cost or at the lower of cost or market value, disclosure of the market value is required.

Exercises

1. Define the following terms relating to the accounting for marketable equity securities: a) Equity securities
b) Valuation allowance
c) Carrying amount
d) Realised gain or loss
e) Net unrealized gain or loss
- 2,

CHAPTER FOUR RECEIVABLES

Objectives

By the end of this lesson, you should be able to: Define receivables and identify the different types of receivables Explain accounting issues related to recognition of accounts receivable Explain accounting issues related to valuation of accounts receivable. Explain accounting issues related to recognition of notes receivable

Introduction

Receivables arise from transactions such as; sale of goods and services, loans made, claims for income tax refunds, claims resulting from mitigation and amounts due from leasing of assets. Receivables from customers normally represent a substantial portion of business enterprise's current assets. Poor screening of applications for credit or an inefficient collection policy may result in large losses. Consequently strong accounting policies must be put in place.

Classification of receivables

Receivables are broadly classified into trade receivables and non-trade receivables.

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• Trade receivables

These arise from the normal operating activities of the business, that is, the credit sale of goods and services to customers. These receivables may be evidenced by a formal written promise to pay in which case they are referred to as Notes Receivable. In most cases however trade receivables are unsecured open accounts often referred to as simply Accounts Receivable.

• Non Trade receivables

These include all other types of receivables. They arise from a variety of transactions such as: Sale of securities or property other than goods and services, advances to stockholders or directors, claims from losses or damages and deposits with creditors.

Non trade receivables should be summarised appropriately and recorded separately in the financial statements.

Accounts Receivable Valuation

Short-term receivables should be valued and reported at the net realisable value - the net amount expected to be received in cash, which is not necessarily the amount legally receivable. Determining net realisable value however requires an estimation of both uncollectible receivables and any returns or allowances to be granted.

Uncollectible Accounts Receivable:

Sales on any basis other than for cash make possible the subsequent failure to collect the account. An uncollectible account receivable is a loss of revenue that requires, through proper entry in the accounts, a decrease in the asset, accounts receivable and a related decrease in income and stockholders' equity. The loss in revenue and the decrease in income are recognised by recording bad debt expense.

Methods for Recording Uncollectibles:

There are two main methods of recording uncollectible accounts receivable.

- (i) Direct write-off method
- (ii) Allowance method

Direct write-off method

Under this method, no entry is made until a specific account has definitely been established as uncollectible. The loss is then recorded by

Dr. Bad Debt Expense Account

Cr. Accounts Receivable

The bad debt is indeed recorded in the year it is determined that a specific receivable cannot be collected. This method is theoretically deficient because it usually does not match costs with revenues of the period, nor does it result in receivables being stated at estimated net realisable value on the balance sheets. As a result, its use is not considered appropriate, except when the amount uncollectible is immaterial.

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Allowance Method

Under this method, an estimate is made of the expected uncollectible accounts from all sales made on account or from the total of outstanding receivables. This estimate is entered as an expense and an indirect reduction in accounts receivable (via an increase in the allowance account) in the period in which the sale is recorded. Advocates of this method believe that bad debt expense should be recorded in the same period as sale to obtain a proper matching of expenses and revenues and to achieve a proper carrying value for accounts to receivable. The percentage of receivables that will not be collected can be predicted from past experiences, present market conditions, and an analysis of the outstanding balances.

This method is appropriate in situations where it is probable that an asset has been impaired and that the amount of the loss can be reasonably estimated. A receivable is a prospective cash inflow and the probability of its collection must be considered in valuing this inflow.

Notes Receivable Recognition

of Notes receivable:

A note receivable is supported by a formal promissory note, a written promise to pay a certain sum of money at a specific future date. Such a note is a negotiable instrument that is signed by a maker (drawer) in favour of designated payee who may legally and readily sell or otherwise transfer the note to others. Notes are classified as either interest bearing or non-interest bearing. (Zero-interest bearing notes).

Interest bearing notes have a stated rate of interest whereas zero-interest-bearing notes include interest as part of their face amount instead of stating it explicitly.

Valuation of Notes receivable:

Short-term notes are generally recorded at face value (less allowance) because the interest implicit in the maturity value is immaterial. A general rule is that notes treated as cash equivalents (maturities of 3 months or less) are not subject to premium or discount amortisation. Long-term Notes receivable, however, should be recorded and reported at the present value of the cash expected to be collected. When the interest stated on interest bearing note is equal to the effective (market) rate of interest, the note sells at face value.

When the stated rate is different from the market rate, the cash exchanged (present value) is different from the face value of the note. The difference between the face value and the cash exchanged, either a discount or a premium, is recorded and amortised over the life of a note to approximate the effective (market) interest rate.

1. Notes issued at face value:

Western Company lends Nairobi Imports Ltd. Kshs1, 000, 000 in exchange for a Kshs 2,000,000, 3-year note bearing interest at 10% annually. The market rate of interest for a note of similar risk is also 10%.

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The present value or exchange price of the note is computed as follows:

	Kshs	Kshs
Face value of the note		2,000,000
Present value of the principal:		
Kshs 2,000,000 (PVF _{3, 10%}) = Kshs 2,000,000(0.75132)	1,502,640	
Present value of the interest		
200,000 (PVFA _{3, 10%}) = Kshs 200,000 (2.48685)	<u>497,370</u>	
		<u>(2,000,000)</u>
Difference		
0		

Recording

	Kshs	Kshs
Notes Receivable	2,000,000	
Cash		2,000,000

Western Company would recognise the interest earned each year as follows:

	Kshs	Kshs
Cash	200,000	
Interest Revenue		200,000

2. Notes not issued at Face Value:

(a) Zero-Interest Bearing Notes

If a zero interest-bearing note is received solely for cash, its present value is the cash paid to the issuer. Because, both the future amount and the present value of the note are known the interest rate can be computed, that is, it is implied. The implicit rate is the rate that equates the cash paid with the amount receivable in the future. The difference between the future (face) amount and the present value (cash paid) is recorded as discount and amortised to interest revenue over the life of the note.

Illustration

Jamii Company receives a 3-year Kshs. 1,000,000 zero interest bearing note, the present value of which is Kshs. 772,180.

- Calculate the implicit interest rate.
- Record the transaction.

$$PV = FV (PVIF_{1\%, 3})$$

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$$772,180 = 1,000,000 (\text{PVIF}_{r\%, 3})$$

$$\text{PVIF}_{r\%, 3} = \frac{772,180}{1,000,000} = 0.77218$$

Checking from the present value tables, the implicit rate is found to be 9%.

The entry to record the transaction is as follows:-

	Kshs	Kshs
Notes Receivable	1,000,000	
Discount on Notes Receivable (1,000,000-772,180)		227,820
Cash		772,180

Note: The Discount on Notes Receivable is a valuation account and is reported on the Balance sheet as a contra asset account to notes receivables. The discount is then amortised and interest revenue is recognised annually using the effective interest method.

The 3-year Discount amortisation and interest revenue schedule is shown as follows:

Schedule of Note Discount Amortisation Effective interest Method 0% Note Discounted at 9%

	I Cash Received <u>Kshs</u>	II Interest Revenue <u>Kshs</u>	III Discount Amortised <u>Kshs</u>	IV Carrying Amount of Note at year end <u>Kshs</u>
Date of Issue				772,180
End of Year 1	-0-	69,496	69,496	841,676
End of Year 2	-0-	75,751	75,751	917,427
End of Year 3	<u>-0- 1,000,000</u>	<u>82,573</u>	<u>82,573</u>	
	<u><u>-0-</u></u>	<u><u>227,800</u></u>	<u><u>227,800</u></u>	

Note:

- Interest Revenue (column II)= carrying amount (IV) * 9%
- Discount amortised (III) =Interest Revenue - Cash Received
- Carrying Amount of Note at the end of the year = carrying amount at the beginning + Discount amortised in the year.

The Interest Revenue at the end of the first year using the effective interest method is recorded as follows.

	<u>Kshs</u>	<u>Kshs</u>
Discount on Notes receivable	69,496	
Interest Revenue		69,496

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Note

The entries for the other periods are made in a similar manner. The amount of the discount, Kshs 227,820 in this case represents the interest revenue to be received from the note over the three years.

Activity

Make the necessary entries for end of year 2 and 3.

(a) Interest bearing notes:

Often the stated rate and the effective rate are different. The note may be issued at a discount or premium.

• Notes issued at a discount Illustration

Johnson Company made a loan to Weldon Company and received in exchange a 3-year Kshs 1,000,000 note bearing interest of 10% annually. The market rate of interest for a note of similar risk is 12%. Show the necessary calculations and recordings.

Solution

The exchange price of the note is calculated as follows.

	<u>Kshs</u>	<u>Kshs</u>
Face value of the note		1,000,000
Present value of the principal: $1,000,000 (PVIF_{3,12\%})$	711,780	
Present value of the interest: $100,000 (PVIFA_{3,12\%})$	240,183	
Total Present value of the note	(951,963)	
Difference (Discount)	<u>48,037</u>	

The note was exchanged at a discount in this case, due to the fact that the effective rate of interest (12%) is greater than the stated rate (10%).

The receipt of the note by Johnson Co. is recorded as follows:

	<u>Kshs</u>	<u>Kshs</u>
Notes receivable	1,000,000	
Discount on Notes Receivable		48,037
Cash		951,963

The discount is the amortised and interest revenue recognised annually as follows:

Schedule of Note Discount Amortisation Effective interest Method 10% Note Discounted at 12%

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	Cash Received <u>Kshs</u>	Interest Revenue <u>Kshs</u>	Discount Amortised <u>Kshs</u>	Carrying Amount of Note <u>Kshs</u>
Date of Issue				951,963
End of Year 1	100,000	114,236	14,236	966,199
End of Year 2	100,000	115,944	15,944	982,143
End of Year 3	<u>100,000</u>	<u>117,857</u>	<u>17,857</u>	1,000,000
	<u>300,000</u>	<u>348,037</u>	<u>48,037</u>	

At the end of Year 1, Johnson Co. receives Kshs 100,000 in cash. But its interest revenue is Kshs 114,236. The difference (114,236 - 100,000) is the amortised discount of Kshs 14,236.

The receipt of the annual interest and amortisation of the discount for the first year are recorded by Johnson as follows:- (As per amortisation schedule)

	<u>Kshs</u>	<u>Kshs</u>
Cash	100,000	
Discount on Notes Receivable	14,236	
Interest Revenue		114,236

Note:

The entries for the other years are recorded in the same manner.

Activity

Make the necessary entries for end of year 2 and 3.

• Notes Issued at a Premium

When the present value exceeds the face value, the note is exchanged at a premium. The premium is recorded as a debit and amortised using the effective interest method over the life of the note as annual reductions in the amount of interest revenue recognised. **Illustration**

Johnson Company made a loan to Weldon Company and received in exchange a 3-year, Kshs 1,000,000 note bearing interest at 12% annually. The market rate of interest for a note of similar risk is 10%. Show the necessary calculations and recordings.

Solution

The exchange price of the note is calculated as follows:

	<u>Kshs</u>	<u>Kshs</u>
Face value of the note		1,000,000
Present value of the principal: 1,000,000 (PVF _{3,10%})	751,315	
Present value of the interest:	<u>298,422</u>	<u>1,049,737</u>

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120,000 (PVIF_{A3,10%}) Total

present value

Difference (Premium)

49,737

The Note is exchanged at a premium since the stated rate is higher than the effective rate of interest.

The premium is then amortised and interest revenue recognised annually as follows:

Schedule of Note Premium Amortisation Effective interest Method 12% Discounted at 10%

	Cash Received Kshs	Interest Revenue Kshs	Premium Amortised Kshs	Carrying Amount of Note at year end Kshs
At date of issue				1,049,737
End of year 1	120,000	104,974	15,026	1,034,711
End of year 2	120,000	103,741	16,259	1,018,452
End of year 3 Less rounding off error	120,000	101,845	18,155	1,000,297 297 1,000,000

At the end of year1, the company receives cash amounting to Kshs 120,000 but its interest revenue is Kshs 104,974. The difference of Kshs 15,026 represents the premium amortised.

The receipt of the annual interest and amortisation of the premium for the first year are recorded by Johnson as follows: - (As per amortisation schedule)

	<u>Kshs</u>	<u>Kshs</u>
Cash	120,000	
Premium on Notes Receivable		15,026
Interest Revenue		104,974

Note:

The entries for the other years are recorded in the same manner.

Activity

Make the necessary entries for end of year 2 and 3.

Sale of receivables

Receivables can be passed on to a third party either without recourse or with recourse

• Sale without re-course

The purpose of selling receivables without re-course is to shift to the purchase of the receivables the risk of those losses, the effort of collection and the waiting period that result from granting of credit cards. When merchants accepts credit cards they avoid accounting costs and doubtful

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accounts expenses and may obtain cash immediately in return for a 4 to 7% free discount. The amount of the fee depends on the volume of the business a merchant generates for the credit card Company

• Sale with recourse

When receivables are sold with recourse, the seller (transferee) in effect guarantees the receivables. The (purchaser) is reimbursed of failure of debtors to pay the full amount anticipated at the time of sale e.g. a sale of Kshs 19,500 of trade receivables with a face amount of Kshs 21,000 and a carrying amount of Kshs 20,400 is recorded as follows

	Dr	Cr.
	Kshs	Kshs
Cash	19,500	
Allowances for	600	
Loss on transfer	900	
Trade accounts receivables		21,000

Assignment of receivables

Instead of selling receivables, a business enterprise may borrow money using the receivables. This may involve a pledge of the receivables under a contract providing that the proceeds from the collection of the receivables must be used to retire the loan. Alternatively receivables may be assigned under a more formal arrangement whereby a borrower (assignor) pledges the receivables to a lender (assignee) and signs a promissory note payable. Assignment gives the assignee the same right to bring action to collect the receivables that the assignor possesses. The assignor retains the credit risks and continuous collection efforts and promises to make good any receivables that cannot be collected.

Example

In January A Ltd assigned accounts receivables of Shs 50,000 to F Limited. F Ltd remitted 90% of receivables less 2% fee. The entries involved are;

	Dr	Cr.
	Kshs	Kshs
Assigned accounts receivable	50,000	
Accounts receivable a/c		50,000
Cash	44,100	
Interest expenses	900	
Notes payable		45,000

Jan 31st collected Shs 30,150 and paid this amount to F Ltd. including interests at 1% a month or Shs 45,000 unpaid balance of loan

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Balance on January 31st are assigned accounts receivables equal to Shs 19,850 i.e. (Shs 50,000-30,150)

Notes payable to F limited = Shs 15,300 i.e. (45,000-29,700)

	Dr.	Cr.
	Shs	Shs
Cash	30,150	
Assigned a/c receivable		30,150
Notes payable to F	29,700	
Interest expenses	450	
Cash		30,150

On Feb 28th he collected Shs 17,000 and paid balance owed to F limited plus interest at 1% per month on Shs 15,300 unpaid balance of loan.

	Dr.	Cr.
	Shs	Shs
Cash	17,000	
Assigned a/c receivable		17,000
Notes payable F	15,300	
Interest expenses	153	
Cash		453

Feb 25th

Transferred balance of assigned receivables to the accounts receivable ledger accounts

Dr accounts receivables	2,850	
Cr. assigned a/cs receivables		2,850

Exercises

1. What is an aging of accounts receivables schedule. Describe how such an analysis may be used to estimate doubtful accounts' expense and to analyse the quality of accounts receivable.
2. Some accountants classify Doubtful Accounts Expense as an operating expense, while others classify it as a contra revenue account. Discuss the reasoning behind these alternative positions.
3. The following accounts appear in the general ledger of Delphis Company at the end of year 2002.

	Kshs
Sales	2,400,000
Accounts receivable	1,000,000
Allowance for doubtful accounts	4,000 Dr

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Prepare a journal entry to recognise doubtful accounts expense for each independent assumption below:

- The allowance for doubtful accounts is increased to a balance of Kshs30,000
- The company recognises 2% of sales as doubtful accounts expense
- Through an aging schedule of the accounts, Kshs 49,000 of accounts' receivable is estimated to be uncollectible.

- You are auditing the financial statements of Kwale Company at the end of its financial year. Your review of accounts receivable and discussions with client personnel disclose that the following items are included in the accounts receivable (both controlling and subsidiary ledgers):

	Kshs
Customers' accounts with credit balances	7,900
Receivables from officers	22,500
Advances to employees	4,200
Customers' accounts known to be uncollectible	5,880

Prepare a correcting journal entry to reclassify items that are not trade accounts receivable and write off the uncollectible accounts' receivable.

Exercises

- What is the distinction between trade receivables and miscellaneous receivables? Give two examples of each.
- At what point should receivables be recorded?
- Describe two methods of accounting for cash (sales) discounts.
- What is an aging of accounts receivable? Describe how such an analysis may be used to estimate doubtful accounts expense and to analyze the quality of accounts receivable.
- Explain the distinction between factoring and assigning of accounts receivable.
- The following accounts appear in the general ledger of Beckman company at the end of the current year:

Sales	sh.2,400,000
Accounts receivable	1,000,000
Allowance for doubtful accounts	4,000DR

Prepare a journal entry to recognize doubtful accounts expense for each independent assumption below:

- The allowance for doubtful accounts is increased to a balance of sh. 30,000.
- The company recognizes 3% of sales as doubtful accounts expense.
- Through the aging of the accounts , sh. 45,500 of accounts receivable is estimated to be uncollectible.

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CHAPTER FIVE INVENTORIES

Objectives

By the end of this lesson, you should be able to: Distinguish between perpetual and periodic inventory systems. Describe and compare the cost flow assumptions used in accounting for inventories. Identify the reasons why a given inventory method is selected.

Introduction

Inventories consist of goods held for sale to customers, partially completed goods, and material and supplies to be used in production. Inventory items are acquired and sold continuously by a merchandising enterprise; or acquired, placed in production, converted to a finished product, and sold by a manufacturing enterprise. The sale of merchandise or finished products is the primary source of revenue for most non-service business enterprises. In retail or merchandising operation, inventories consist principally of products purchased for resale in their existing form. A manufacturing enterprise on the other hand has several types of inventories: materials, parts and factory supplies; goods in process (work in progress); and finished goods.

Inventory Procedures:

Two methods may be employed to maintain inventory records:

- (i) Periodic inventory system
- (ii) Perpetual inventory system

Perpetual Inventory System:

Under this system, a continuous record of changes in inventory is maintained in the inventory account. That is, all purchases and sales (issues) of goods are recorded directly in the inventory account as they occur. The accounting features of a perpetual inventory system are:

- Purchases of merchandise for resale or raw materials for production are debited to inventory rather than to purchases.
- Carriage inwards, purchase returns and allowances, and purchase discounts are recorded in inventory rather than in separate accounts.
- Cost of goods sold is recognised for each sale by debiting the account, cost of goods sold, and crediting inventory.
- Inventory in a control account that is supported by a subsidiary ledger of individual inventory records. The subsidiary records show the quantity and cost of each type of inventory on hand.

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The perpetual inventory system provides a continuous record of the balances in both the inventory account and the cost of goods sold account. Under a computerized record keeping system, additions to and issuance from inventory can be recorded nearly instantaneously. **Periodic Inventory System:**

Under this system, the quantity of inventory on hand is determined only periodically. All acquisitions of inventory during the accounting period are recorded by debits to a purchases account. The total in the purchases account at the end of the accounting period is added to the cost of the inventory on hand at the beginning of the period to determine the total cost of the goods available for sale during the period. Ending inventory is subtracted from the cost of goods available for sale to compute the cost of goods sold. Under the periodic inventory system, the cost of goods sold is a residual amount that is a dependent upon a physically counted ending inventory.

Illustration:

Beginning inventory	200 units @ shs 60 = shs 12,000
Purchases	800 units @ shs 60 = shs 48,000
Sales	600 units @ shs 90 = shs 54,000
Ending inventory	400 units @ Kshs 60 = Kshs 24,000

Show the entries to record the above transactions under both inventory systems.

Solution

(i) Perpetual inventory system

Beginning inventory, 200 units at Kshs 60

The inventory account shows the inventory on hand at Kshs 12,000

Purchase 800 units at Kshs 60

Inventory account	Accounts payable	48,000	48,000
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Sale of 600 units at Kshs 90

Sales sh. 54,000

Cost of goods sold (600 at Kshs 60)	36,000
-------------------------------------	--------

End of period entries for inventory accounts, 400 units at Kshs 60. No entry necessary. The inventory account, shows the ending balance of Kshs 24,000 (Kshs 12,000 + Kshs 48,000 - Kshs 36,000).

(ii) Periodic Inventory System:

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Beginning inventory, 200 units at Kshs 60	The inventory account	shows		
the inventory on hand at Kshs 12,000				
Purchase 800 units at Kshs 60	Purchases account	Accounts payable	48,000	48,000
Sale of 600 units at Kshs 90	Accounts receivable	Sales	54,000	No 54,000
entry is made as far as cost of goods sold is concerned.				
End of period entries for inventory accounts, 400 units at Kshs 60.			24,000	48,000
Inventory (ending by count)	Cost of goods sold		36,000	12,000
Purchases	Inventory (Beginning)			

Inventory Valuation Methods:

A major objective of accounting for inventories is the proper determination of income through the process of matching appropriate costs against revenues. Cost for inventory purposes may be determined under any one of several assumptions as to the flow of cost factors; the major objective in selecting a method should be to choose one, which, under the circumstances, most clearly reflects periodic income. The most widely used methods of inventory valuation are:-

- First In, First Out Method (FIFO)

- Last In, First Out Method (LIFO)
- Weighted Average Method

Other special inventory valuation methods include:-

- Retail method
- Gross profit method
- Lower of cost or market value

First In, First Out (FIFO):

This method assumes a flow of costs based on the assumption that the oldest goods on hand are sold first. This assumption generally conforms to reality; management usually finds it desirable to keep the oldest goods moving out to customers in order to keep fresh or new goods on hand. The inventory remaining must therefore represent the most recent purchases. In all cases where this method is used, the ending inventory and cost of good sold would be the same at the end of the month whether a perpetual or periodic system is used. This is true because the same costs will always be first in and, therefore, first out whether cost of goods sold is computed as goods are sold throughout the accounting period (perpetual system) or as a residual at the end of the accounting period (periodic system).

Advantages of FIFO:

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- The ending inventory is close to current cost. Because the first goods in are the first goods out, the ending inventory amount will be composed of the most recent purchases. This approach generally provides a reasonable approximation of replacement cost on the balance sheet when process changes have not occurred since the most recent purchases.
- When the physical flow of goods is actually first in- first out, the FIFO method closely approximates specific identification method. At the same time it does not permit manipulation of income because the enterprise is not free to pick a certain cost item to be charged to expense.

Disadvantages of FIFO:

Current costs do not match against current revenues on the income statement. The oldest costs are charged against the more current revenue, which can lead to distortions in gross profit and net income.

Illustration:

The following transactions relate to Musongari Company, for the month of February 2004.

<u>Date</u>	<u>Purchases</u>	<u>Sold (Issued)</u>	<u>Balance</u>
2/2/2004	1000 units @ Kshs 10		1000 units
5/2/2004	5000 units @ Kshs 12		6000 units
9/2/2004		3000 units	3000 units
15/2/2004	4000 units @ Kshs 14		7000 units
20/2/2004		1000 units	6000 units
24/2/2004		2000 units	4000 units
26/2/2004		3000 units	1000 units
28/2/2004	5000 units @ Kshs 13		6000 units

Determine the cost of goods sold during the month and the ending inventory as at the end of the month of February using:

- Perpetual Inventory System
- Periodic Inventory System

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Solution FIFO -Perpetual Inventory

Date	Number of units value	Purchase units sold value	Number of units Kshs purchased Kshs	Sales	Balance No. of (Issued) Kshs	Value
2/2/2004	1000@ Kshs 10	10,000			1000 @ Kshs 10	10,000
5/2/2004	10 5000@ Kshs 12	60,000			1000 @ Kshs 10 5000 @ Kshs 12	70,000
9/2/2004			3000: 1000 @ Kshs 10 2000@ Kshs 12	34,000	3000@ Kshs 12	36,000
15/2/2004	4000@ Kshs 14	56,000			7000 3000@ Kshs 12 4000@ Kshs 14	92,000
20/2/2004			1000 @12	12,000	6000 :2000@12 4000@14	80,000
24/2/2004			2000 @12	24,000	4000 @14	56,000
26/2/2004			3000 @ 14	42,000	1000 @ 14	14,000
28/2/2004	5000@ Kshs 13	65,000			6000: 1000 @ 14 5000@ Kshs 13	79,000

FIFO-Periodic inventory

Using this method, the cost of the ending inventory is computed by taking the cost of the most recent purchases and working back until all units in the inventory are accounted for.

Date	Number of units	Unit cost (Kshs)	Total cost (Kshs)
28/02/2004	5000	13	65,000
15/02/2004	<u>1000</u>	14	<u>14,000</u>
Ending inventory	<u>6000</u>		<u>79,000</u>

Note

In all cases where FIFO is used, the inventory and cost of goods sold would be the same at the end of the month whether a perpetual or periodic system is used. This is true because the same costs will always be first in and therefore first out whether cost of goods sold is computed as goods are sold throughout the accounting period (perpetual) or as a residual at the end of the accounting period (periodic).

Last In First Out (LIFO)

This method assumes a flow of inventory costs based on the assumption that the most recently acquired goods on hand are sold first. Under this view, the latest costs are most closely associated with current revenue, and thus the matching principle of income determination is carried out.

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Illustration

Using the data pertaining to Musongari Co. for the month of February, Determine the cost of goods sold during the month and the ending inventory as at the end of the month of February using: (a) Perpetual Inventory System (b) Periodic Inventory System

Date	Number of units purchased	Purchase value Shs	Number of units sold (issued)	Sales value Shs	Number of units remaining (Balance)	Value Shs
2/2/04	1,000@shs 10	10,000			1,000	10,000
5/2/04	5,000@ Shs12	60,000			6,000	70,000
9/2/04			3,000@shs 12	36,000	3,000	34,000
15/2/04	4,000@shs 14	56,000			7,000	90,000
20/2/04			1,000@shs 14	14,000	6,000	76,000
24/2/04			2,000@shs 14	28,000	4,000	48,000
26/2/04			3,000(1000 @shs14, 2000@shs12	36,000	1000	12,000
28/2/04	5,000@shs 13	65,000			6,000	77,000

Exercises

1. There are two systems of maintaining inventory records: (a) periodic inventory system and (b) perpetual inventory system. What are the basic differences between the two systems, and under what circumstances should each be used?
2. What costs should be included in the cost of inventories? What objectives are considered in deciding what costs are to be included in inventories?
3. Differentiate between the weighted average method and the moving weighted average of determining the cost of inventories.
4. The Simple company sells one product , Simplot. Presented below is information for January for the Simple company.

Jan. 1 Inventory
 Jan. 4 Sale
 Jan. 11 Purchase
 Jan. 13 Sale
 Jan. 20 Purchase
 Jan. 27 Sale

100 units at sh. 5 each
 80 units at sh. 8 each
 150 units at sh. 6 each
 120 units at sh. 8.50 each
 150 units at sh. 7 each
 100 units at sh. 9 each

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A physical count on January 31 shows that ending inventory is 100 units. The company use the periodic method.

Required:

A) Compute

i) Cost of ending inventory

ii) Gross profit

Under a) LIFO cost flow assumption

b) FIFO cost flow assumption

c) weighted average method

B) Assuming the company uses the perpetual system repeat A above.

Courtesy of Rodgers Wesonga

CHAPTER SIX SPECIAL VALUATION METHODS

Objectives

By the end of this lesson, you should be able to: Determine ending inventory by applying the gross profit method and the retail inventory method. Explain and apply the lower of cost or market rate Identify when inventories are valued at net realisable value

Introduction

Some odd circumstances require the use of special valuation methods. For instance, where stock has been lost through fire and a physical count is not possible or where there is a large departmental store and monthly inventory figures are needed and monthly counts are not feasible. Such situations call for the development and use of estimation techniques to value the end inventories without undertaking a physical count. The gross profit method and the retail method are widely used as estimation methods.

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Retail method

The retail method of estimating the cost of inventories is used primarily by retailing enterprises. Under the periodic inventory system, the cost of the ending inventories is subtracted from the total cost of goods available for sale to compute the cost of goods sold. Under the retail method, a record of goods available for sale at selling prices is kept separate from the accounting records, and sales for the accounting period are deducted from this total to determine the ending inventories at selling prices. The ending inventories valued at selling prices then are reduced to estimated cost by multiplying the inventories at selling prices by the cost percentage computed for the accounting period.

Uses of the retail method

- To verify the reasonableness of the cost of inventories at the end of the accounting period. By using a different set of data from that used in pricing inventories, accountants may establish that the valuation of inventories is reasonable.
- To estimate the cost of inventories for interim accounting periods and for income tax purposes.
- To permit the valuation of inventories when selling prices are the only available data.

The use of this method allows management to mark only the selling prices on the merchandise and eliminates the need for reference to specific purchase invoices.

Special items relating to retail method

The retail inventory method becomes more complicated when such items as freight costs are treated as part of the cost of purchase.

- Purchase return and allowance are ordinarily considered as a reduction of the cost price and the retail price.
- Purchase discounts are usually considered as a reduction of purchase.
- Sales return and allowances are considered to be proper adjustments to gross sales.
- Normal shortage (breakage, damage, and theft) should reduce the retail column because these goods are no longer available for sale. This amount is not considered in computing the cost to retail percentage but it shown as a reduction similar to sales to arrive at ending inventories at retail.
- Abnormal shortage should be deducted from both the cost and retail column, prior to calculating the cost to retail ratio and reported as a special inventory amount or as a loss.
- Companies often provide employees with special discounted to encourage loyalty. This should be deducted from the retail in the same way as sales. This discount should not be considered in the cost to retail percentage because they do not reflect an overall change of the selling price.

Although the retail method enables estimation of the value of inventories without a physical count of the items on hand, the accountant should insist that a physical inventory be taken periodically.

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Otherwise, shrinkage due to shoplifting, breakage, and other causes might go undetected and might result in an increasingly overstated inventories valuation.

Normal shrinkage in the inventories may be estimated on the basis of the goods that were available for sale. The method frequently used is to develop a percentage from the experience of past years, such as 2% of the retail value of goods available for sale. This percentage is used to determine the estimated shrinkage, which is deducted, together with sales, from goods available for sale at retail prices to compute the estimated inventories at retail prices. The cost of normal shrinkage is included in the cost of goods sold; the cost of abnormal shrinkage (theft, unusual spoilage, etc) that is material in amount is reported separately in the income statement.

The estimated cost of the inventories is computed by use of a cost percentage, that is, the relationship between the cost of goods available for sale during an accounting period and their retail value. The reliability of this procedure rests on the conditions that

- A uniform relationship exists between selling price and cost for all goods available for sale during the. When sales are made to employees or selected customers at a special discount price, such discounts are added to sales to compute the estimated inventories at retail prices period
- if the markup on individual inventory items differs, the distribution of items in the ending inventories is roughly the same as the “mix” in the total goods available for sale during the period. When one of these conditions is not present, the accuracy of the retail method is improved by applying it to the individual departments of the business enterprise, and adding the resulting departmental inventories to compute the estimated cost of the total inventories.

Retailing Terminology

The following terms are used in the application of the retail method of estimating the cost of inventories:

- Original selling price -The price at which goods originally are offered for sale.
- Markup -The initial margin between the selling price and cost. It also is referred to as gross margin or mark-on.
- Additional markup - An increase above the original selling price.
- Markup cancellation - A reduction in the selling price after there has been an additional markup. The reduction does not reduce the selling price below the original selling price. Additional markups less markup cancellations are referred to as net markups.
- Markdown - A reduction in selling price below the original selling price.
- Markdown cancellation - An increase in the selling price, following a markdown, which does not increase the new selling price above the original selling price. Markdowns less markdown cancellations are referred to as net markdowns.

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Illustration

Assume that an item that cost Kshs 200 is priced to sell for Kshs 300 a unit. The markup is Kshs 100 (50% of cost or $\frac{100}{300} \times 100 = 33\frac{1}{3}\%$ of selling price). In response to strong demand for the item an additional markup of Kshs30 is added, so that the selling price is increased to Kshs 330. As the demand slackens, the price is reduced to Kshs 310 by a markup cancellation of Kshs 20. Subsequently, in order to dispose of the remaining units, the selling price is reduced to Kshs 250 by a markup cancellation of Kshs 10 and a markdown of Kshs 50. Finally, if management concludes that the remaining units will be sold at a price of Kshs 280, a markdown cancellation of Kshs 30 is required to increase the selling price from Kshs 250 to Kshs 280.

Retail Method - Valuation at Average Cost

Under this method the cost percentage is determined after net markups are added to and net markups deducted from the goods available for sale. This method is however accurate only if the goods on hand (ending inventory) consist of a representative sample of all goods available for sale during the year.

Illustration

Assume the following data pertaining to Kanyange wholesalers for the year ending 31st December 2003. Calculate the cost value of the ending inventory using the retail-average cost method.

	Cost	Retail
	Kshs	Kshs
Beginning inventory	31,620	54,000
Net purchases	150,380	220,000
Additional mark up		10,000
Mark up cancellation		(4,000)
Mark downs		(21,750)
Mark down cancellations		1,750
Net sales		(180,000)
Ending inventory at retail		<u>80,000</u>

Solution

	Cost	Retail
	Kshs	Kshs
Beginning inventory	31,620	54,000
Net purchases	150,380	220,000
Net markups (10,000-4000)		6,000
Less: Net markdowns (21,750-1,750)		<u>(20,000)</u>
Goods available for sale	<u>182,000</u>	260,000
Cost percentage (182,000/260,000) =70%		
Less net sales		<u>(180,000)</u>
Ending inventory at retail		<u>80,000</u>

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Ending inventory at cost (80,000*70%) **Shs 56,000**

Retail Method - Valuation at Lower of Average Cost or Market

The retail method may be adapted to produce inventory valuations approximating the lower of average cost or market when there have been changes in the costs and selling prices of goods during the accounting period. The crucial factor in the estimate of cost of the ending inventory by the retail method is the treatment of net markups and net markdowns in the computation of the cost percentage. The inclusion of net markups and the exclusion of net markdowns in the computation of the cost percentage produce an inventory valued at the lower of average cost or market. This is sometimes called the conventional retail method

Illustration

Using the previous data relating to Kanyange wholesalers for the year ending 31st December 2003, calculate the cost value of the ending inventory using the retail-lower of average cost or market method.

Solution Cost	Retail	Kshs	Kshs
Beginning inventory		31,620	54,000
Net purchases		150,380	220,000
Net markups (10,000-4000)			<u>6,000</u>
Goods available for sale		<u>182,000</u>	280,000
Cost percentage (182,000/280,000) =65%			
Less: Net sales			(180,000)
Less Net markdowns (21,750-1,750)			<u>(20,000)</u>
Ending inventory at retail			<u>80,000</u>
Ending inventory at cost (80,000*65%)		<u>Shs 52,000</u>	

Retail Method Valuation at Last-In, First-Out

If the last-in, first-out method is used to estimate the cost of inventories, the conventional retail method must be modified. The retail method may be adapted to approximate LIFO cost of the ending inventories by the computation of a cost percentage for purchases of the current accounting period only. The objective is to estimate the cost of any increase (LIFO layer) in inventories during the accounting period.

Because LIFO is a cost (not lower-of-cost-or-market) method of inventory valuation, both net markups and net markdowns are included in the computation of the cost percentage for purchases of the current period.

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Illustration

Using the data pertaining to Kanyange wholesalers for the year ending 31st December 2002, calculate the cost value of the ending inventory using the retail-LIFO method

Solution

	Cost	Retail
	Kshs	Kshs
Beginning inventory	31,620	54,000
Net purchases	150,380	220,000
Net markups (10,000-4000)		6,000
Less: Net markdowns (21,750-1,750)		<u>(20,000)</u>
Goods available for sale	<u>182,000</u>	260,000
Less net sales		<u>(180,000)</u>
Ending inventory at retail		<u>80,000</u>
Cost percentage for net purchases, including Markups and net markdowns (Shs 150,380/206,000*)=73%		

*Shs 206,000=Shs 220,000+6,000-20,000

Net mark up = 10,000-2,000= Shs 6,000

Net mark down = 21,750-1,750 =Shs20, 000

LIFO layer

	Cost	Retail
	Kshs	Kshs
Beginning inventory at cost	31,620	54,000
Added layer (0.73*26,000)	<u>18,980</u>	<u>26,000</u>
Ending inventory at retail		<u>80,000</u>
Ending inventory at LIFO cost	<u>50,600</u>	

Under retail LIFO procedures, computation of a cost percentage for current year's purchases is required only when an increase in inventories (at retail) occurs during the current year. The cost percentage is computed for the sole purpose of pricing the incremental layer in inventories. In contrast, if a decrease (LIFO liquidation) in inventories takes place, the ending inventories consist of a fraction of the beginning inventory cost.

Retail Method - Valuation at First-In, First-Out

Gross profit method

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The gross profit method is useful for several purposes:

- To control and verify the validity of inventory cost;
- To estimate interim inventory valuations between physical counts; and
- To estimate the inventory cost when necessary information normally used is lost or unavailable.

The procedure involved is one of reducing sales to a cost basis; that is, cost of goods sold is estimated. The estimated cost of goods sold then is subtracted from the cost of goods available for sale to compute the estimated cost of the ending inventories.

In the event that both merchandise and inventory records are destroyed by fire, the inventory cost may be estimated by use of the gross profit method. The gross profit and cost of goods sold percentages are obtained from prior years' financial statements, which presumably are available. The beginning inventories amount for the current year is the ending inventory amount of the preceding year. Net purchases are estimated from copies of the paid checks returned by the bank and through correspondence with suppliers. Sales are computed by reference to cash deposits and by an estimate of the outstanding accounts receivable through correspondence with customers.

Illustration

Assume the following data for Dexter Corporation for Year 2002.

	Kshs
Beginning inventory at cost	500,000
Net purchases	2,000,000
Net sales	3,000,000

The company's normal gross profit rate is 40% of net sales. Using the above facts, calculate the estimated ending inventory.

Solution

	Kshs
Beginning inventory	500,000
Net purchases	<u>2,000,000</u>
Cost of goods available for sale	2,500,000
Less estimated ending inventory (X)	<u>700,000</u>
Cost of goods sold (60%* net sales)	<u>1,800,000</u>

Estimated ending inventory (X) = cost of goods available for sale - cost of goods sold

The cost of the ending inventories estimated by the gross profit method is reasonably consistent with the usual method of valuing inventories. This follows from the fact that the gross profit percentage is based on historical records that reflect the particular method of valuing the inventories. If the inventories are valued at LIFO, the estimated inventories will approximate LIFO cost; therefore, if the gross profit method is used as a basis for recovering an insured fire loss, the inventories should be restated for insurance purposes to current fair value at the time of the fire.

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Calculation of gross profit percentage

Sometimes the gross profit percentage is stated as a percentage of cost. In such situations the gross profit percentage must be restated to a percentage of net sales to compute the cost percentage (based on net sales) for the period. The following formulas may be used to express the relationship.

- $$\text{Percentage mark-up on selling price} = \frac{\% \text{ mark-up on cost}}{100\% + \% \text{ mark-up on cost}}$$
- $$\text{Percentage mark-up on cost} = \frac{\% \text{ mark-up on selling price}}{100\% - \% \text{ mark-up on selling price}}$$

For example, if the gross profit is stated as 25% of cost, the gross profit percentage may be restated to 20% of net sales as follows:

$$\begin{aligned}\text{Gross profit as a \% of net sales} &= \frac{25\%}{100\% - 25\%} \\ &= 20\%\end{aligned}$$

When 20% is subtracted from 100%, the difference 80% represents the cost percentage of net sales.

Valuation of Inventories at Replacement Costs

The valuation of inventories at replacement costs has been advocated by accountants who believe that the current asset section of the balance sheet should reflect current fair values. The cost methods of inventory valuation frequently understate the value of inventories, particularly during periods of rising prices. The significance of replacement costs as a measure of inventory value varies considerably depending on the type of inventories involved. In the retail market, the selling prices of staple commodities, such as sugar, copper, cotton, etc., tend to follow cost prices closely. In such situations, replacement costs of inventories are important to management and outsiders.

Replacement-cost valuation in the preparation of financial statements is not generally accepted. Perhaps the closest practical approach is the first-in, first-out method. Unless prices are rising rapidly, the FIFO method of pricing presents inventories in the balance sheet at or near current replacement costs without a departure from the cost principle. The need for disclosure of replacement or current costs of inventories arises when the last-in, first-out method is used for pricing inventories.

Valuation of Inventories at Net Selling Prices

The valuation of inventories at net selling prices (sales prices less direct costs of completion and disposal) has some appeal, especially when one considers that economic value is added as the goods are brought to market. For example, in a retail store, goods are more valuable than they were at the wholesaler's warehouse; value is added by the process of bringing the goods nearer the ultimate market. In a manufacturing enterprise, costs are blended together, and a product emerges that is more valuable than the sum of the production costs. However, this method of inventory valuation has not been widely adopted for two reasons:

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- The lack of objectivity in determining the net selling prices.
- The fact that the selling price has not been realized in cash or cash equivalents.

Accountants generally consider revenue to be realized at the time of sale of goods, not at the time of production. The valuation of inventories at net selling prices is appropriate for some types of business enterprises producing commodities that have readily determinable market prices. When the production of such commodities is complete, revenue may be considered realized. In some enterprises having selling prices established by contract, the sale is reasonably assured, and completed inventories may be valued at net selling prices.

Exercises on inventories

- 1) The following information is taken from the accounting records of Broadways Company limited for the current year.

	Cost	Retail
	Kshs	Kshs
Beginning inventory	25,000	46,200
Purchases (net)	120,000	191,800
Net markups 12,000 Sales (net)	180,000	
Net markdowns		3,800

You are to assume that all net markdowns apply to purchases of the current year, and that it is appropriate to treat the entire inventory as a single department, with no markdowns having occurred during the prior year.

- (a) Compute the ending inventory at lower of average cost or market by the retail method (b) Compute the ending inventory using the retail LIFO method

1. Describe the application of the retail method when cost is determined on a last in first out basis.
2. Describe the computation of the cost percentage when inventory is determined at average cost by the retail method.
3. List three uses that may be made of the gross profit method.
4. Explain the possible limitations of the use of an average cost percentage for prior years to estimate inventories by the gross profit method.
5. Shaka Limited uses the gross profit method to estimate monthly inventories. In recent months gross profit has averaged 30% of net sales. The following data are available for the month of December 2003.

	Kshs
Inventories, December 1	53,160
Purchases	240,000
Purchases returns	10,000
Freight-in	12,000
Gross sales	338,000
Sales returns and allowances	20,000

Compute the estimated cost of the inventories on 31st December 2003 using the gross profit method

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Exercises

1. List the major uses of the gross profit method.
2. For what purposes may the retail method of inventory valuation be used?
3. What is the basic assumption as to the composition of the ending inventory when the retail method is applied on the basis of average cost or on the basis of the lower of average cost or market?

Courtesy of Rodgers Wesonga

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CHAPTER SEVEN PROPERTY, PLANT AND EQUIPMENT (IAS 16)

Objectives

By the end of this lesson, you should be able to : To describe the major characteristics of property, plant and equipment. Identify the costs included in the initial valuation of property, plant and equipment Understand accounting issues related to acquiring and valuing plant assets Describe the accounting treatment for costs subsequent to acquisition

Introduction

Property, plant and equipment are tangible assets that are held by an enterprise for use in the production or supply of goods or services, for rental to others, or for administrative purposes and are expected to be used during more than one period. International Accounting Standard (IAS) 16 states that an item of property, plant and equipment should be recognised as an asset when:

- It is probable that future economic benefits associated with the asset will flow to the enterprise,
- the cost of the asset to the enterprise can be measured reliably.

In determining whether an item satisfies the first condition for recognition, an enterprise needs to assess the degree of certainty attaching to the flow of future economic benefits on the basis of the available evidence at the time of initial recognition. Existence of sufficient certainty that the future economic benefits will flow to the enterprise necessitates an assurance that the enterprise will receive rewards attaching to the asset and will undertake the associated risks. This assurance is usually only available when the risks and rewards have passed to the enterprise.

The second criterion for recognition is usually readily satisfied because the exchange transaction evidencing the purchase of the asset identifies its cost. In the case of a self constructed asset, a reliable measurement of the cost can be made from the transactions with parties external to the enterprise for the acquisition of the materials, labour and other inputs used during the construction process.

Classifying plant assets

Plant assets are often sub divided into four classes:

- Land, such as a building site
- Land improvements, such as driveways, parking lots, fences and underground sprinkler systems.
- Buildings, such as stores, factories, offices and warehouses.
- Equipment, such as cash registers, coolers, factory machinery and delivery equipment.

Determining the cost of plant assets

An item of property, plant and equipment that qualifies as an asset should initially be measured at its cost. Cost consists of all expenditure necessary to acquire the asset and make it ready for its intended use.

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Components of Cost

The cost normally comprises: purchase price, including import duties and non refundable purchase taxes, and any directly attributable costs of bringing the asset to working condition for its intended use; any trade discounts and rebates are deducted in arriving at the purchase price. Examples of directly attributable costs include; cost of site preparation, initial delivery and handling costs, installation costs, and professional fees such as for architects and engineers.

Note

Administration and other general overhead costs are not a component of the cost of the asset unless they can be directly attributed to the acquisition of the asset or bringing the to its working condition. Similarly, start up and pre production costs do not form part of the cost of an asset unless they are necessary to bring the asset to its working condition.

The cost of a self constructed asset is determined using the same principles as an acquired asset, however the cost of abnormal amounts of wasted material, labour or other resources incurred in the production of a self constructed asset is not included in the cost of the asset.

Land

The cost of land includes:

- cash purchase price
- closing costs such as title and attorney's fees
- real estate brokers' commissions
- accrued property taxes and other liens on the land assumed by the purchaser
- When vacant land is acquired, these costs include expenditures for clearing, draining, filling and grading.
- Where the land has an old building that needs to be removed to make the site suitable for construction of a new building, all demolition and removal costs less any proceeds from salvaged materials. **Illustration**

Castle Brewing Company acquires real estate at a cash cost of Kshs 100M. The property contains an old warehouse that is razed at a net cost of Kshs 0.5M (Kshs 700,000 less Kshs 250,000 proceeds from salvaged materials). Additional expenditures consist of the lawyer's fee, Kshs 1M and the real estates broker's commission, Kshs 10M. Compute the cost of the land that should be recorded in the company's balance sheet.

Solution

	Land
	Kshs
Cash price of property	100,000,000
Net removal cost of warehouse	500,000
Lawyer's fee	1,000,000
Real estate broker's commission	<u>10,000,000</u>
Cost of land	<u>111,500,000</u>

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Land improvements

The cost of land improvements includes all expenditures necessary to make the improvements ready for their intended use. For example, the cost of a new company parking lot will include the amount paid for paving, fencing, and lighting. These improvements have limited useful lives and their maintenance and replacement are the responsibility of the company. Thus, these costs are debited to Land Improvements Account and not the lands account and are depreciated over their useful lives.

Buildings

All necessary expenditures relating to the purchase or construction of a building are charged to the buildings account. When a building is purchased, such costs include:

- The purchase price,
- Closing costs such as lawyer's fee, title insurance
- Costs to make the building ready for its intended use such as expenditure; for remodelling rooms and offices, repairing the roof, floors, electrical wiring and plumbing.

When a company constructs a plant such as the East African Breweries plant at Ruaraka, cost consists of the contract price plus payments made by the owner for architects' fees, building permits, and excavation costs. In addition interest costs incurred to finance the project are included in the cost of the asset when a significant period of time is required to get the asset ready for use. However the inclusion of interest is limited to the construction period. When construction has been completed, subsequent interest payments on funds borrowed to finance the construction are debited to interest expense account.

Equipment

The cost of equipment consists of; the cash purchase price, sales taxes, freight charges, and insurance during transit paid by the purchaser. It also includes expenditures required in assembling, installing, and testing the unit. However motor vehicle licences and accident insurance on company trucks and cars are expensed as incurred, because they represent annual recurring expenditures and do not benefit future periods.

Illustration

East African Portland Cement Company purchases a limestone grinding machine from Japan at a cash price of Kshs 80M. In shipping the machinery the company incurred Kshs 5M and Kshs 4M as freight charges and transit insurance respectively. Value added tax charged at the port of entry amounted to Kshs 15M. The cost of installing and testing the machine amounted to Kshs 2M. Compute the cost of the machine that ought to be reflected in the company's balance sheet.

Solution	Factory Machinery
	Kshs (Millions)
Cash price	80
Freight charges	5
Transit insurance	4
Value added tax	15
Installation and testing	2

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Cost of factory machine 106

The summary entry to record the purchase and related expenditure is:

Dr. Factory Machinery	Kshs106M	
Cr. Cash/Bank		Kshs 106M

Lump-Sum purchase

A lump-sum purchase occurs when more than one type of asset is acquired in a single transaction. In such a case, the single lump-sum purchase price must be allocated equitably to the individual components. The most common method of allocation is based on the relative fair market values of the individual assets.

Illustration

Continental Hotel purchases a family owned motel at the Kenyan coast at a cash cost of Kshs 75M on 1st October 2003. The individual asset components comprised of; Land, Buildings and Equipment. The fair market values of these assets are; Land Kshs 25M, Buildings Kshs 40M and Equipment Kshs 35M. Using the fair market value basis, allocate the purchase price.

Solution

<u>Asset class</u>	<u>Fair market Value</u>	<u>% of total fair Market value</u>	<u>Computation</u> %*Purchase price	<u>Cost allocation</u>
	Kshs (M)			Kshs (M)
Land	25	25/100*100 (25%)	25%*75	18.75
Buildings	40	40/100*100 (40%)	40%*75	30
Equipment	35	35/100*100 (35%)	35%*75	26.25
	100			75

The entry to record the lump-sum purchase is;

Dr	Land	Kshs (M)	Kshs (M)
	Buildings	18.75	
	Equipment	30	
		26.25	
	Cr. Cash		75

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Note: Normally the book values of the individual assets on the seller's books are not used since they are rarely indicative of fair market values at the date of purchase.

Borrowing costs

IAS 23 states that borrowing costs that are directly attributable to the acquisition, construction and production of a qualifying asset are those costs which would have been avoided if the expenditure on the qualifying asset had not been made. Such cost should be included in the cost of the asset when it is probable that they will result in future economic benefits to the enterprise and the costs can be measured reliably. To the extent that funds are borrowed specifically for the purpose of obtaining a qualifying asset, the amount of borrowing costs eligible for capitalisation on that asset should be determined as the actual borrowing costs incurred on that borrowing during the period less any investment income on the temporary investment of those borrowings.

Borrowing costs may include:

- Interest on bank overdrafts and short term /long term borrowings
- Amortisation of discounts or premiums relating to borrowings
- Amortisation of ancillary costs incurred in connection with the arrangement of borrowings
- Exchange differences arising from foreign currency borrowings to the extent that they are regarded as an adjustment to interest costs.

A qualifying asset is defined as an asset that necessarily takes a substantial period of time to get ready for its intended use or sale.

Exchange of Asset

An item of property, plant and equipment may be acquired in exchange or part exchange for a dissimilar item of property, plant and equipment or other asset. The cost of such an item is measured at the fair value of the asset received, which is equivalent to the fair value of the asset given up adjusted by the amount of any cash or cash equivalents transferred. Such an item of property, plant and equipment may be acquired in exchange for a similar asset that has a similar use in the same line of business and which has a similar fair value. An item of property, plant and equipment may also be sold in exchange for an equity interest in a similar asset. In both cases, since the earnings process is incomplete, no gain or loss is recognized on the transaction. The cost of the new asset in this case is the carrying amount of the asset given up.

Escrow Statement

Purchase to a real estate transaction generally engage (bank or escrow company) to handle details of the transaction. When the transaction is closed each party receives an escrow statement showing complete details of the transaction. The escrow statement shows items such as the selling price of property mortgage note assumed by the acquirer transfer taxes commission charged to seller escrow fees cash received from the acquirer and any amount paid to either party to complete the transaction.

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Subsequent expenditure

Subsequent expenditure relating assets that have already been recognized should be added to the carrying amount of the asset when it is probable that future economic benefits, in excess of the originally assessed standard of performance of the existing asset, will flow to the enterprise. All other subsequent expenditure should be recognized as an expense in the period in which it is incurred. Such expenditure is only recognized as an asset when the expenditure improves the condition of the asset beyond its originally assessed standard of performance. Examples of improvements, which result in, increased future economic benefits include:

- Modification of an item of plant to extend its useful life, including an increase in its capacity.
- Upgrading machine parts to achieve a substantial improvement in the quality of output.
- Adoption of new production processes enabling a substantial reduction in previously assessed operating costs.

Exercises

1. On April 1, 2002 Highway development company acquired real estate, on which they planned to construct a small office building, by paying Kshs 15M in cash. An old warehouse on the property was razed down at a cost of Kshs 100,000; the salvaged materials were sold for Kshs 75,000. Additional expenditures before construction began included Kshs 150,000 lawyer's fee for work concerning the land purchase, Kshs 120,000 real estate broker's fee, Kshs 95,000 architect's fee, and Kshs 285,000 to put in drive ways and a parking lot. (a) Determine the amount to be reported as the cost of the land.

(b) For each cost not used in part (a), indicate the account to be debited

2. Plant asset acquisitions for selected companies are as follows;

- (a) Standard Newspapers company limited purchased an ultra modern state of the art printing press with the capability of printing 500, 000 copies per hour for Kshs 300M. In addition the company paid Kshs 25M as shipping charges, Kshs 20M as insurance during transit, Kshs 75M as import taxes and Kshs 50M for installation. The company made the following entry for the printing press;

	Kshs (Millions)	Kshs (Millions)
Printing press	425	
Freight in	25	
Insurance expense	20	
Bank		470

- (b) Roy Transporters Company Limited purchased a delivery truck for Kshs 17M. In addition it paid Kshs 3.5M for import duties, Kshs 0.5M for the first year's insurance, Kshs 75,000 motor vehicle license, and Kshs 25,000 for painting the company logo and name on the truck. The truck was recorded as follows:

	Kshs	Kshs
Delivery truck	17,500,000	
Licenses and taxes Expense	3,575,000	
Miscellaneous Expense	25,000	
Bank		21,100,000
Required		

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Prepare the correct entries that should have been made for each acquisition above.

3. Czar Company Limited makes a lump sum purchase of plant assets at a total cost of Kshs 60M. On the seller's books, land was recorded at cost, Kshs 5M and the book values of the buildings and equipment were Kshs 15M and Kshs 10M respectively. At the time of the purchase the assets were valued as follows: land Kshs. 20M, buildings Kshs. 35M and equipment Kshs. 10M.
- Required
- Prepare the journal entry to record the acquisition.

CHAPTER EIGHT INTANGIBLE ASSETS (IAS 38)

Objectives

By the end of this lesson, you should be able to: Describe the characteristics of intangible assets Identify the types of intangible assets Identify the costs included in the initial valuation Explain the accounting issues related to impairments

Introduction

The basic characteristic that distinguishes intangible assets from tangible assets is that the former are not physical in nature. In legal terminology the distinction is maintained consistently, the term intangibles being applied to all non-physical properties, including cash, accounts and notes receivable, and investments in corporate securities. However, in accounting terminology intangible assets do include patents, copyrights, trademarks, trade names, secret formulas, organisation costs, franchises, licenses, and goodwill (the excess of cost of an acquired business enterprise over the current fair value of identifiable net assets acquired).

One reason for distinguishing between tangible and intangible asset is that it often is difficult to identify intangible assets. Evidence of the existence of intangible assets may be vague, and the relationship between expenditure and the emergence of an asset may be difficult to establish objectively. The economic value of both tangible and intangible assets is dependent on their ability to generate future revenue and earnings, and this often is as difficult to measure for tangible assets as it is for intangibles. However, mere physical existence (obsolete machinery, for example) is no guarantee of economic value, nor does the absence of physical existence (the Listerine formula, for example) preclude economic value. For some business enterprises, the value of intangible assets may exceed the value of the tangible assets.

Cost of intangible assets

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A business enterprise may acquire intangible assets from others, or it may develop internally certain types of intangible assets. The general objectives in accounting for intangible assets are comparable to those for tangible assets; the initial cost is determined and charged against the revenue, which these assets help to generate. A significant and permanent decline in the value of intangible asset is debited to expenses in the year the decline occurs. Generally, such write-offs are not reported as extraordinary losses in the income statement.

When an intangible asset is purchased, its cost can be measured with little difficulty. It may be necessary to estimate the value of non-monetary cost among various assets acquired as a group. The principles used in dealing with these problems, as previously described in relation to plant assets, are equally applicable to intangible assets.

The Board concludes that a company should record as assets the costs of intangible assets acquired from other enterprises or individuals. Costs of developing, maintaining, or restoring intangible assets which are not specifically identifiable, have indeterminate lives, or are inherent in a continuing business and related to an enterprise as a whole-such as goodwill-should be deducted from income when incurred.

Intangible assets acquired singly should be recorded at cost at date of acquisition. Cost is measured by the amount of cash disbursed, the fair value of other assets distributed, the present value of amounts to be paid for liabilities incurred, or the fair value of consideration received for stock issued ...

Intangible assets acquired as part of a group of assets or as part of an acquired company should also be recorded at cost at date of acquisition. Costs is measured differently for specifically identifiable intangible assets is an lacking specific identification. The cost of the identifiable assets is an assigned part of the total cost of the group of assets or enterprise acquired, normally based on the fair value of the individual assets. The cost of unidentifiable intangible assets is measured by the difference between the cost of the group of assets or enterprise acquired and the sum of the assigned costs of individual tangible and identifiable intangible assets acquired less liabilities assumed. Cost should be assigned to all specifically identifiable intangible assets; cost of identifiable assets should not be included in goodwill.

Identifiable intangible assets

Certain intangible assets, such as patents, copyrights, and franchises, can be identified as distinct and separable property rights; others, such as goodwill, are difficult to identify. The more common identifiable intangible assets are discussed as below:

Patents

A patent is a grant by the government giving the owner the exclusive right to manufacture and sell a particular invention for a period of 17 years. Patent rights may be assigned in part or in their entirety. Frequently, contracts require payments of royalties to the owner of the patent for the right

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to use or manufacture a patented product. Legally, patents cannot be renewed, but in practice their effective life often is extended by obtaining patents on slight variations and improvements near the end of the legal life of the original patent.

A patent has economic value only if the protection it affords against competition results in increased earnings through an ability to operate at a lower cost, to manufacture and sell a product, or to obtain a higher price for goods and services. The economic life of a patent generally is much shorter than its legal life; therefore, amortisation over the period of usefulness usually is necessary.

If a patent is purchased, its cost is measured by the purchase price plus any incidental costs. The purchase of a patent from another party is recorded as follows:

A patent does not include automatic protection against infringement; owners must prosecute those who attempt to infringe their patents and defend against infringement suits brought by owners of similar patents. The cost of successfully establishing the legal validity of a patent should be capitalised, because such cost will benefit revenue over remaining economic life of patent. However, a patent infringement suit may take years to resolve, and the accounting treatment of legal cost during this period must recognise the uncertainties involved by expensing such costs. If the legal decision is favourable, legal costs may be paid by the losing party; if the legal decision is adverse, both the cost of the infringement suit and the unamortised cost of the patent should be written off, because no further economic benefits are expected to result from the patent.

The right to use a patent that is owned by another party under a licensing contract is not recorded as an intangible asset, unless a lump-sum payment is made at the outset of such a contract. The periodic royalty payments are recorded as factory overhead or as operating expense, depending on the use made of the patent.

If a patent is developed as a result of a business enterprise's research and development efforts, the cost assigned to the patent includes only the research and development costs incurred internally, because all such costs must be expensed as incurred.

Copyrights

A copyright is a grant by the government giving an author, creator, or artist the exclusive right to publish, sell, or otherwise control literary or artistic products for the life of the author plus 50 years. The rights granted under copyrights may be acquired by paying royalties, by purchase, or by obtaining a copyright on a product developed by a business enterprise. The problems that arise in measuring the cost of copyrights are comparable to those already discussed in connection with patents.

Although a copyright has a long legal life, its economic life is limited to the period of time for which a commercial market exists for the publication. In order to achieve a proper matching of costs and revenue, copyright costs are amortised against the total revenue that is anticipated from the copyright. Because of the difficulty encountered in estimating copyright revenue and because experience indicates that such revenue generally results over only a few years, copyrights typically are amortised over relatively short periods of time.

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Licenses and contracts

Many business enterprise invest considerable sums of capital to obtain licenses to engage in certain types of business activities or to acquire rights to use copyrighted materials owned by others. For example, network affiliation contracts and film rights probably are the most valuable assets of enterprise engaged in the broadcasting industry. Without an FCC license, it would be impossible for a broadcaster to earn revenue; a network-affiliated station is more valuable than an independent station because of network-supplied programming; the rights to show old movies are an important source of revenue for television broadcasters.

The cost of license or a contract is recorded in the accounting records and is amortised over the accounting periods expected to benefit. An license generally is amortised over the period of 40 years; a network-affiliation contract is amortised over the period specified in the contract; and the film rights purchased by a television station generally are amortised on an accelerated basis because first showing generates more advertising revenue than reruns. If a license or a contract is cancelled or for any reason becomes worthless, any unamortised cost should be written off.

Trademarks, trade names, and secret formulas

Trademarks, trade names, secret formulas, and various distinctive labels are important means of building and holding customer acceptance for certain products. The value of such product identification and differentiation stems from the ability of the business enterprise to sell products in large volume and at prices higher than those of unbranded products.

Trademarks, trade names secret formulas, and labels are property rights that can be leased, assigned, or sold. Their economic lives continue as long as they are used, and their cost is amortised over their estimated economic lives or 40 years, whichever is shorter.

The value of trademarks, trade names, or secret formulas often is enhanced as the enterprise succeeds in building consumer confidence in the quality of product distributed under a particular brand name. Presumably this growth in value is not without cost, because enterprises spend large sums for advertising and otherwise promoting trade names. The relationship between promotional expenditures and the increase in the value of trade names is nebulous; therefore, accountants do not assign a cost to this intangible asset, except when it is acquired by purchase.

Organisation costs

The organisation of a corporate business enterprise usually requires a considerable amount of time, effort, and cost. Compensation must be paid to those who conceive, investigate, and promote the idea; legal fees relating to drafting of corporate charter and bylaws, accounting fees, and incorporation fees are incurred; and costs may be incurred in conducting initial meetings of stockholders and directors. All these expenditures are made with the expectation that they will contribute to future revenue. It is clear, therefore, that the cost of organising a corporate enterprise

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logically should be treated as an asset and not as an expense. On the other hand, items such as losses incurred in the early years of a corporate enterprise, bond discount and issuance costs, large initial advertising expenditure, or discount on capital stock, are not included in organisation costs. Expenditures incurred in connection with the issuance of shares of capital stock, such as professional fees and printing costs, generally are deducted from the proceeds received for the stock. Similar expenditure relating to the issuance of bonds or mortgages notes payable is deferred and is amortised over the term of the debt.

Theoretically, organisation costs have an economic life as long as the corporate enterprise remains a going concern. Because the life of the most corporate enterprises is unlimited, organisation costs may be viewed as a permanent asset that will continue in existence until the enterprise goes out of business. Despite the logic of this position, organisation costs generally are amortised over a five-year period. However, amortisation over a maximum period of 40 years is permitted by generally accepted accounting principles.

Franchises

A franchise is a right or privilege received by a business enterprise for the exclusive right to engage in business in a certain geographic area. The franchise may be acquired from a governmental unit or from another enterprise. For example, public utilities generally receive a franchise from the state and are subject to specific regulations. A retailer may obtain an exclusive right from a manufacturer to sell certain products in a specified territory; an operator of a restaurant may obtain the right to utilise trade names and recipes developed by another enterprise for instance Steers, Nandos, McDonalds and Southern Fried Chicken.

Some franchises that are granted by manufacturers or retail chains (franchisers) might cost substantial amounts. The amount paid for such a franchise is recorded by the franchisee as an intangible asset and amortised over its expected economic life. The proceeds received by franchisers are recorded as revenue when the contractual commitments to franchisees are fulfilled. If the right to operate under a franchise is limited to 10 years, for example, the amortisation period should not exceed 10 years. Although some franchises prove to be worthless within a short period of time, others may increase substantially in value if the location and product prove successful.

Leasehold costs

The purchase of an existing lease right and a lump-sum payment to acquire rights to explore for oil and minerals on land are valuable property rights which frequently are included under intangible assets in the balance sheet. However, because such assets represent rights to use tangible assets, they may be included under plant assets.

Research and development costs

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Definition of Research and Development

Research is original and planned investigation undertaken with the prospect of gaining new scientific or technical knowledge and understanding.

Development is the application of research findings or other knowledge to a plan or design for the production of new or substantially improved materials, devices, products, and processes, systems or services to the commencement of commercial production or use.

Research Activities

- Activities aimed at obtaining new knowledge
- Search for application of research findings or other knowledge
- Search for products or process alternatives
- Formulation and design of possible new or improved product or process alternatives

Development activities

- Evaluation of product or process alternatives
- The design, construction and testing of pre- production prototypes and models
- The design of tools, jigs, moulds and dies involving new technology
- The design, construction and operation of a pilot plant that is not of a scale economically feasible for commercial production.

Activities closely associated with R&D but are neither research nor development

- Quality control during commercial production, including routine testing of products
- Seasonal or other periodic design changes to existing products
- Routine efforts to refine, enrich or otherwise improve upon the qualities of an existing product
- Engineering follow-through in an early phase of commercial production.
- Adaptation of an existing capability to a particular requirement or customer's need as part of a continuing commercial activity
- Activities, including design and construction engineering, related to the construction, relocation, rearrangement, or start-up of facilities or equipment other than facilities or equipment used solely for a particular research and development project.

Components of research and development costs

Research and development should comprise all costs that are directly attributable to research and development activities or that can be allocated on a reasonable basis to such activities. They include, when applicable:

- Salaries, wages and other employment related costs of personnel engaged in research and development activities.
- Costs of materials and services consumed in research and development activities.
- Depreciation of property, plant and equipment to the extent that these assets are used for R&D activities
- Overhead costs, other than general administrative costs, related to R&D activities.

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- Other costs, such as the amortization of patents and licenses, to the extent that these assets are used for R&D activities

Recognition of research and development costs.

The allocation of R&D costs to different periods is determined by the relationship between the costs and the economic benefits that the enterprise expects to derive from the R&D activities. When it is probable that the costs will give rise to future economic benefits and the costs can be measured reliably, the costs qualify for recognition as an asset

Research costs

These should be recognized as an expense in the period in which they are incurred and should not be recognized as an asset in a subsequent period. This is because the nature of research is such that there is insufficient certainty that future economic benefits will be realized as a result of specific research expenditures.

Development costs

The development costs of a project should be recognized as an asset when all of the following criteria are met:

- The product or process is clearly defined and the costs attributable to the product or process can be separately identified and measured reliably.
- The technical feasibility of the product or process can be demonstrated
- The enterprise intends to produce and market, or use, the product or process
- The existence of a market for the product or process or, if it is to be used internally rather than sold, its usefulness to the enterprise can be demonstrated.
- Adequate resources exist, or their availability can be demonstrated, to complete the project and the market or use the product or process.

The development cost of a project recognised as an asset should not exceed the amount that is probable of being recovered from related future economics benefits, after deducting further development costs, related production cost and selling and administrative costs directly incurred in the product.

The development cost of a project should be written down to the extent that the unamortised balance, taken together with further development cost, related production cost and selling and administration cost directly incurred in marketing the product is no longer probable of being recovered from the expected future economic benefits. The unamortised balance of development cost of a project should be written off as soon as any of the criteria in paragraph 17 for the recognition of the development cost as an asset cease to be met. The amount of write-down should be recognised as an expense in the period in which the write down or write off occurs. The unamortised balance of development costs of a project recognised as an asset is reviewed at the end each period. Current circumstances or events may indicate that the unamortised balance, taken together with the other relevant costs, exceeds the related future economic benefits. Alternatively, the unamortised balance may no longer meet the criteria for recognition as an asset.

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Cost of an internally generated intangible asset

The cost of internally generated intangible asset comprises all expenditure that can be directly attributed, or allocated on a reasonable and consistent basis, to creating, producing and preparing the asset for its intended use. The cost includes, if applicable:

- Expenditure on Materials and services used or consumed in generating the asset.
- Salaries, wages and other employment related costs of personnel directly engaged in the generation process.
- Any expenditure that is directly attributable to generating the asset such as fees to register a legal right and the amortization of patents and licenses that are used to generate the asset
- Overheads that are necessary to generate the asset and that can be allocated on a reasonable and consistent basis to the asset for instance an allocation of the depreciation of property, plant and equipment.

Note

The following are not components of the cost of an internally generated intangible asset:

- Selling, administrative and other general overhead expenditure unless this expenditure can be directly attributed to preparing the asset for use.
- Clearly identified inefficiencies and initial operating losses incurred before an asset achieves planned performance
- Expenditure on training staff to operate the asset.

Subsequent expenditure

Subsequent expenditure on an intangible asset after its purchase or its completion should be recognised as an expense when it is incurred unless:

- It is probable that this expenditure will enable the asset to generate future economic benefits in excess of its originally assessed standard of performance.
- This expenditure can be measured and attributed to the asset reliably

When the above conditions are met, the subsequent expenditure should be added to the cost of the intangible asset.

Identifiable intangible assets

Certain intangible assets, such as patents, copyrights, and franchises, can be identified as distinct and separable property rights; others, such as goodwill, are difficult to identify. The more common identifiable intangible assets are discussed in the following sections.

Patents

A patent is a grant by the federal government giving the owner the exclusive right to manufacture and sell a particular invention for a period of 17 years. Patent rights may be assigned in part or in

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their entirety. Frequently, contracts require payments of royalties to the owner of the patent for the right to use or manufacture a patented product. Legally, patents cannot be renewed, but in practice their effective life often is extended by obtaining patents on slight variations and improvements near the end of the legal life of the original patent.

A patent has economic value only if the protection it affords against competition results in increased earnings through an ability to operate at a lower cost, to manufacture and sell a product, or to obtain a higher price for goods and services. The economic life of a patent generally is much shorter than its legal life; therefore, amortization over the period of usefulness usually is necessary.

If a patent is purchased, its cost is measured by the purchase price plus any incidental costs. The purchase of a patent from another party is recorded as follows:

A patent does not include automatic protection against infringement; owners must prosecute those who attempt to infringe their patents and defend against infringement suits brought by owners of similar patents. The cost of successfully establishing the legal validity of a patent should be capitalized, because such cost will benefit revenue over remaining economic life of patent. However, a patent infringement suit may take years to resolve, and the accounting treatment of legal cost during this period must recognize the uncertainties involved by expensing such costs. If the legal decision is favorable, legal costs may be paid by the losing party; if the legal decision is adverse, both the cost of the infringement suit and the unmortized cost of the patent should be written off, because no further economic benefits are expected to result from the patent.

The right to use a patent owned by other under a licensing contract is not recorded as an intangible asset, unless a lump-sum payment is made at the outset of such a contract. The periodic royalty payments are recorded as factory overhead or as operating expense, depending on the use made of the patent.

If a patent is developed as a result of a business enterprise's research and development efforts, the cost assigned to the patent includes only the research and development costs incurred internally are assigned to the patent, because all such costs must be expensed as incurred. Accounting for research and development costs is covered in another section of this chapter.

Copyrights

A copyright is a grant by the federal government giving an author, creator, or artist the exclusive right to publish, sell, or otherwise control literary or artistic products for the life of the author plus 50 years. The rights granted under copyrights may be acquired by paying royalties, by purchase, or by obtaining a copyright on a product developed by a business enterprise. The problems that arise in measuring the cost of copyrights are comparable to those already discussed in connection with patents.

Although a copyright has a long legal life, its economic life is limited to the period of time for which a commercial market exists for the publication. In order to achieve a proper matching of

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costs and revenue, copyright costs are amortized against the total revenue that is anticipated from the copyright. Because of the difficulty encountered in estimating copyright revenue and because experience indicates that such revenue generally results over only a few years, copyrights typically are amortized over relatively short periods of time. On occasion, copyrights thought be valueless may bounce back to life with renewed vigor. An outstanding example is old movies: Their production and copyright costs previously had been fully amortized, but these films suddenly became extremely valuable with the development of television and the apparent incidence of insomnia among television viewers. However, this increase in the value of copyrights was not reflected in the balance sheets of motion picture producers.

Licenses and contracts

Many business enterprise invest considerable sums of capital to obtain licenses to engage in certain types of business activities or to acquire rights to use copyrighted materials owned by others. For example, Federal Communications Commission (FCC) licenses, network affiliation contracts, and film rights probably are the most valuable assets of enterprise engaged in the broadcasting industry. Without an FCC license, it would be impossible for a broadcaster to earn revenue; a network-affiliated station is more valuable than an independent station because of network-supplied programming; the right s to show old movies are an important source of revenue for television broadcasters.

The cost of license or a contract is recorded in the accounting records and is amortized over the accounting periods expected to benefit. An FCC license generally is amortized over the period of 40 years; a network-affiliation contract is amortized over the period specified I the contract; and the film rights purchased by a television station generally are amortized on an accelerated basis because first showing generate more advertising revenue that reruns.⁵ If a license or a contract is canceled or for any reason becomes worthless, any unamortized cost should be written off.

Trademarks, trade names, and secret formulas

Trademarks, trade names, secret formulas, and various distinctive labels are important means of building and holding customer acceptance for certain products. The value of such product identification and differentiation stems from the ability of the business enterprise to sell products in large volume and at prices higher than those for unbranded products.

Trademarks, trade names secret formulas, and labels are property rights that can be leased, assigned, or sold. Their economic lives continue as long as they are used, and their cost is amortized over their estimated economic lives or 40 years, whichever is shorter.

The value of trademarks, trade names, or secret formulas often is enhanced as the enterprise succeeds in building consumer confidence in the quality of product distributed under a particular brand name. Presumably this growth in value is not without cost, because enterprises spend large sums for advertising and otherwise promoting trade names. The relationship between promotional

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expenditures and the increase in the value of trade names is nebulous; therefore, accountants do not assign a cost to this intangible asset, except when it is acquired by purchase.

Organizational costs

The organization of a corporate business enterprise usually requires a considerable amount of time, effort, and cost. Compensation must be paid to those who conceive, investigate, and promote the idea; legal fees relating to drafting of corporate charter and bylaws, accounting fees, and incorporation fees are incurred; and costs may be incurred in conducting initial meetings of stockholders and directors. All these expenditures are made with the expectation that they will contribute to future revenue. It is clear, therefore, that the cost of organizing a corporate enterprise logically should be treated as an asset and not as an expense. On the other hand, items such as losses incurred in the early years of a corporate enterprise, bond discount and issuance costs, large initial advertising expenditure, or discount on capital stock, are not included in organization costs. Expenditures incurred in connection with the issuance of shares of capital stock, such as professional fees and printing costs, generally are deducted from the proceeds received for the stock. Similar expenditure relating to the issuance of bonds or mortgages notes payable are deferred and amortized over the term of the debt.

Theoretically, the cost of organization has an economic life as long as the corporate enterprise remains a going concern. Because the life of the most corporate enterprises is unlimited, organization costs may be viewed as a permanent asset that will continue in existence until the enterprise goes out of business. Despite the logic of this position, organization costs generally are amortized over a five-year period, because the federal income tax law permits amortization over a period of "not less than five years." However, amortization over a maximum period of 40 years is permitted by generally accepted accounting principles.

Franchises

A franchise is a right or privilege received by a business enterprise for the exclusive right to engage in business in a certain geographic area. The franchise may be acquired from a governmental unit or from another enterprise. For example, public utilities generally receive a franchise from state or federal agencies and are subject to specific regulations; a retailer may obtain an exclusive right from a manufacturer to sell certain products in a specified territory; an operator of a restaurant may obtain the right to utilize trade names and recipes developed by another enterprise.

Some franchises granted by manufacturers or retail chains (franchisors) may cost substantial amounts. The amount paid for such a franchise is recorded by the franchisee as an intangible asset and amortized over its expected economic life. The proceeds received by franchisors are recorded as revenue when the contractual commitments to franchisees are fulfilled.⁶ If the right to operate under a franchise is limited to 10 years, for example, the amortization period should not exceed 10 years. Although some franchises prove to be worthless within a short period of time, others may increase substantially in value if the location and product prove successful.

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An operating right issued by the Interstate Commerce Commission or a similar state agency to a motor carrier (trucker) to transport goods with limited competition over specified routes is a form of franchise. Many trucking enterprises acquired such rights to transport goods interstate and paid large sums of money for them. The Motor Carrier Act of 1980 deregulated the interstate trucking industry, thus reducing or eliminating the value of these intangible assets. Consequently, the FASB required that "Unamortized costs of interstate operating rights subject to the provisions of the Act shall be charged to income and, if material, reported as an extraordinary item ... in the income statement."⁷

Leasehold costs

The purchase of an existing lease right and a lump-sum payment to acquire rights to explore for oil and minerals on land are valuable property rights which frequently are included under intangible assets in the balance sheet. However, because such assets represent rights to use tangible assets, they may be included under plant assets.⁸

Unidentifiable intangible assets Goodwill

Thus far we have discussed the major types of identifiable intangible assets. However, the earning power of most prosperous business enterprises is attributable to a variety of factors which cannot be specifically identified, either as tangible assets or as intangible assets. Accountants, business executives, and lawyers often refer to these factors collectively as goodwill.

In ordinary usage of term goodwill is associated with a kindly feeling or benevolence. However, in business and law goodwill has a different meaning. The most acceptable evidence of goodwill is the ability of a business enterprise to earn a rate of return on net assets (owners' investment) in excess of a normal rate for the industry in which the business enterprise operates. Goodwill is the difference between the value of a business enterprise as a whole and the sum of the current fair values of its identifiable tangible and intangible net assets. Goodwill is in essence a "master valuation account" - the missing link that reconciles the current fair value of an enterprise as a going concern with the current fair value of the sum of its parts.

Nature of goodwill

The first obstacle in the path toward and understanding of goodwill is the problem of estimating the current fair value of a business enterprise as a going concern. The current fair value of the enterprise may be greater than the amount of identifiable tangible and intangible assets, because of the presence of unidentifiable intangible assets. A simple example may help to clarify this point. Assume that Parke Company is offered for sale and that the condensed balance sheet below is used as a basis for negotiating a fair price:

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We shall assume that Parke Company is expected to earn an average of \$60,000 a year indefinitely in the future. Because the current fair value of net assets depends directly on their earning power, we can value Parke Company as a going concern, without reference to its balance sheet, by determining the present value of future earnings of \$60,000 a year. A logical way of appraising this is in terms of the rate of return on alternative investment opportunities of comparable risk. We shall assume this rate to be 10%. If it is possible to earn a 10% return on similar investments, the current value of the prospect of receiving \$60,000 a year in perpetuity may be computed by determining the amount which must be invested at 10% to earn an annual return of \$60,000. This procedure is called capitalization of income, and the result in this case is a value for the net assets of Parke Company of \$600,000 ($\$60,000 \div 10\% = \$600,000$), compared with a carrying amount of only \$400,000 ($\$250,000 + \$150,000 = \$400,000$).

If the net assets of Parke Company are worth \$600,000, why are they shown in the balance sheet at only \$400,000? One possibility is that Parke Company's accounting records do not reflect the current fair value of identifiable net assets. Inventories and plant assets, for example, might be worth considerably more than carrying amount, and liabilities might overstated. If these discrepancies are brought to light during the negotiations, appropriate adjustments should be made.

It is possible, however, that the carrying amount of each asset and liability included in the balance sheet closely approximates its current fair value, but still Parke Company's net assets are worth \$200,000 more than carrying amount. Is this accounting exception to the principle that the whole must equal the sum of its parts, or is it simply the case that some of the parts are not included in the balance sheet? The latter is the more likely explanation, and it is apparent that the missing parts are those characteristics of the company that enable it to earn \$60,000 a year (10% of \$600,000) rather than \$40,000 a year (10% of \$400,000). The company apparently has intangible assets that are not included in its balance sheet. Any of the identifiable intangible assets previously discussed in this chapter are possible sources of the unexplained \$200,000 in the current fair value of Parke Company as going concern.

For purposes of this illustration, we shall assume that Parke Company has a patent worth \$50,000 which is not included in the balance sheet, because it was developed internally or because it has been fully amortized. After all identifiable assets, both tangible and intangible, have been appraised, only \$150,000 ($\$200,000 - \$50,000$) remains unexplained, and we have isolated the imputed value of all unidentifiable intangible assets that is, goodwill. Goodwill exists as an asset only because it is impossible to identify separately all sources of the prospective earning power of a business enterprise. This analysis may be summarized as follows for Parke Company:

If patents of \$50,000 and goodwill of \$150,000 were added to the assets of Parke Company, the carrying amount of its net assets would be \$600,000 (assets of \$700,000, less liabilities of \$100,000). Therefore, if the company earned \$60,000 its earnings no longer would be large in relation to the carrying amount of its net assets. Thus, the ability to earn a superior rate of return on net assets which do not include goodwill is evidence that goodwill exists; the ability to earn a normal rate of return on assets which include the goodwill and all identifiable intangible assets is evidence of the existence of goodwill in the amount computed.

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Negative goodwill

Goodwill, as we have defined it, can be positive or negative in amount. Suppose, for example, that the prospective earnings of Parke Company had been estimated at only \$36,000 a year indefinitely into the future and that its identifiable net assets are fairly stated at \$400,000. On a 10% yield basis, the capitalized value of these earnings is \$360,000 ($\$36,000 \div 0.10 = \$360,000$), and it is evident that the carrying amount of the net assets exceeds the current fair value of the company as a whole by \$40,000. This \$40,000 is referred to as negative goodwill.

When the earning potential of a business enterprise is such that the enterprise as a whole is worth less than its net assets, the owners would be better off to dispose of the assets piecemeal, pay the liabilities, and terminate the enterprise. In reality this may not be done because of concern for the welfare of employees, willingness of the owners to continue operating an unprofitable business, optimism about future prospects, or other considerations. Because the presence of negative goodwill suggests that liquidation is the best course of action, positive goodwill is more likely to be found in going concerns than negative goodwill. Although negative goodwill exists in many unsuccessful enterprises, it is not isolated and reported in the balance sheet; the only evidence of its existence is a low rate of return on the net assets of that enterprise.

If an enterprise with negative goodwill is sold as a going concern, the value assigned to the net assets acquired by the purchaser should not exceed the cost actually paid. The total current fair value of identifiable assets acquired less the liabilities assumed occasionally may exceed the price paid for the acquired enterprise. According to APB Opinion No. 16, such an excess over cost should be allocated to reduce the carrying amounts assigned to noncurrent assets (other than long-term investments in marketable securities). If this allocation reduces noncurrent assets to zero, any remaining excess is classified as deferred credit and amortized over a period not exceeding 40 years.⁹

Recording of goodwill (excess of cost over net assets acquired)

The high degree of certainty about the future assumed in the measurement of goodwill of Parke Company, in the example above, does not exist in the real world. Assessing the earnings potential of a business enterprise is an uncertain process, and any resulting estimate of goodwill is a matter of judgment and opinion.

In the face of this uncertainty, accountants have adopted a rule of caution with respect to goodwill. It should be recorded in the accounting records only when its amount is substantiated by an arm's-length transaction. Because goodwill cannot be either sold or acquired separately, accounting recognition of goodwill is restricted to those occasions in which the entire net assets representing a clearly defined segment of a business enterprise, are purchased, and goodwill can be established with reasonable objectivity.¹⁰ In such cases goodwill frequently is labeled as Excess of Cost over Net Assets Acquired.

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Limiting the recording of goodwill to purchased goodwill is admittedly not a perfect solution to the problem. Internally developed goodwill actually may exist in a business enterprise and not be recorded; on the other hand, goodwill acquired in the past may appear in the accounting records when there is no current evidence (in terms of earning power) that it actually exists. The financial statements of business enterprises which have changed ownership will appear to be inconsistent with those of enterprises which have had a continuing existence. For example, assume that Parke Company, which was discussed earlier, has identifiable net assets of \$450,000 and that a new company is formed to purchase its net assets for \$600,000 cash. The opening balance sheet of the new company would include goodwill of \$150,000. Is there any justification for a rule that prohibits the recording of \$150,000 goodwill in the accounting records of Parker Company but permits the inclusion of this amount in the balance sheet of the new company?

On the balance, an affirmative answer is warranted. Specific assets represent resources in which the capital of a business enterprise is invested, to the extent that it has been possible to determine them. The periodic adjustment of these asset valuations by a variable amount labeled “goodwill” to a level consistent with the present value of future earnings not only would be a highly subjective undertaking but also would obscure the significant relationship between actual investment and earning power. If \$150,000 of goodwill had been recorded in the accounting records of Parke Company, not only would there be a serious question as to the validity of this amount, but also the high level of earnings on investment that Parke Company had been able to attain would be concealed. The investment of the new owners, on the other hand, was not \$450,000, but \$600,000. The new owners paid \$150,000 for anticipated earnings in excess of normal, and if only \$150,000 of excess earnings should materialize, this amount will not represent income to the new owners but a recovery of their investment. The position that goodwill should be recognized in the accounting records only when it is evidenced by a purchase transaction appears to be consistent with the cost principle and the basic assumptions underlying the measurement of net income.

Estimate of goodwill

The price to be paid for a business enterprise is established as the result of bargaining between independent parties. The bargaining process includes the possible existence of goodwill. The amount of goodwill to be recorded, however, is determined after the terms of the contract are set by deducting the current fair value of all identifiable net assets from the total purchase price. Accountants are interested in the process of estimating goodwill because they often are called upon to aid in establishing the current fair value of an enterprise, at the time of negotiations for the purchase or sale of an enterprise, and in litigation.

Steps generally required to estimate the current fair value of a business enterprise, and thus the amount of goodwill, are listed on page 539.

¹Estimate the current fair value of all identifiable tangible and intangible assets of the enterprise, and deduct from this total the amount of all liabilities. This gives the current fair value of the identifiable net assets of the enterprise.

²Forecast the average annual earnings that the enterprise expects in future years with the use of its present facilities.

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³Choose an appropriate rate (or rates) of return to estimate the normal annual earnings the enterprise should earn on its identifiable net assets.

⁴Compute the amount of expected annual superior earnings, if any.

⁵Capitalize the expected annual superior earnings, if any, at an appropriate rate (or rates) of return to estimate the present value of such earnings. The present capitalized value of any expected annual superior earnings is the estimated value of goodwill for the enterprise.

In the following sections, an estimate of goodwill is developed for Reed Company (which is for sale) to serve as a basis for a discussion of the problems that arise in connection with each of the five steps listed above.

Estimate of the Current Fair Value of Identifiable Net Assets

Because carrying amounts and current fair values of assets seldom correspond, an appraisal of identifiable assets is necessary to establish the current fair value of the business enterprise (excluding goodwill) and to identify the assets which generate the earnings of the enterprise.

The fair value of current assets, such as cash and accounts receivable, usually will approximate carrying amount. Inventories, if verified by a physical inventory and priced on a fifo or averagecost basis, also may be reasonably stated. Lifo inventories, however, probably are stated in terms of costs incurred many years earlier and should be adjusted to current fair value. The carrying amounts of plant assets are not likely to approximate current fair values. Various methods of indirect valuation may be employed to appraise such assets on a going-concern basis. The current fair value of all identifiable intangible assets should be estimated, even if these assets do not appear in the accounting records. The liabilities of the business enterprise should be reviewed, and any unrecorded liabilities should be estimated and recorded. Liabilities which will not be assumed by the new owners should be ignored, unless payment from present assets is contemplated before the enterprise changes ownership. Assets at current fair values, less liabilities to be assumed by the new owners, gives the adjusted amount (estimated current fair value) of net assets for purposes of estimating the value of goodwill.

The assumed data on top of page 540 for Reed Company illustrate the process of estimating the current fair value of identifiable net assets of a business enterprise.

Forecast of Expected Average Annual Earnings

The aggregate value of any business enterprise depends on its future earnings, not on its past earnings. Thus, the key step in any estimate of the current fair value of an enterprise is a forecast of its future earnings, a process which, unfortunately, can never be more than an intelligent guess. Because the immediate past history of an enterprise ordinarily provides the best available evidence and is most relevant, the usual procedure is to compute the average annual earnings of an enterprise during the past three to six years and to project them into the future, adjusting for any changing conditions that can be foreseen. The estimate of future conditions and earnings generally is made

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by the parties to the transaction and not by accountants. A single year's performance clearly is not a sufficient basis for judgement; on the other hand, little may be gained by reaching too far into the past, because both the internal and the external conditions influencing business operations may have changed radically.

In the compilation of the past earnings record suitable for estimating future earnings, two points should be kept in mind:

It seldom is possible to obtain satisfactory data by simply computing an average of past reported earnings. A more reasonable approach is to work from actual revenue and expenses amounts, because changes in revenue and expenses are likely to be related to projected economic and operating conditions. The effect on earnings of a 10% increase in revenue and a 15% increase in operating expenses, for example, may need to be determined. Past data should be adjusted for changes in the value of assets. For example, if inventories and equipment have been understated in terms of current fair values, adjustments of past cost of goods sold and depreciation expenses must be made. Extraordinary items generally should be omitted from past earnings. In view of the subjectivity of estimates and income measurement, minor adjustments can be ignored.

In the evaluation of an average of past data, particular attention must be given to **significant trends**. For example, two enterprises may have the same five-year average sales, but if the sale of one enterprise have increased in each of the past five years, while the sales of the other have declined steadily, the average sales amounts should be interpreted differently.

An important point, often overlooked in the adjustment of past earnings in the light of future expectations, is that improvements in earnings expected as a result of the efforts of new owners and management should be distinguished carefully from prospective improvements that can be traced to existing conditions. If the purchase of a business enterprise expects to make changes in management, production methods, products, and marketing techniques which will increase earnings in the future, these changes should not be considered in the valuation of the enterprise because they will flow from the efforts of the new owners. Of course, the final price paid for goodwill in any transaction is a matter of bargaining between the purchaser and the seller.

The working paper on pages 542 is a continuation of the Reed Company example. It represents an assumed computation on December 31, Year 10, of estimated future earnings, based upon an average of the results experienced over the past five years. This estimate might be interpreted by the prospective purchaser to indicate a probable range of future annual earnings for Reed Company of, say between \$90,000 and \$120,000 a year. However, for illustrative purposes, we shall use the amount of \$106,000.

Normal Rate of Return

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The rate of return used to capitalize future earnings and to separate superior earnings from ordinary earnings is determined on the basis of the risks and alternatives involved. The objective is to approximate the rate necessary to attract capital to the business enterprise under review, given the existing risk conditions. The cost of capital, as other costs, varies in relation to a variety of factors. The primary cause of difference in the rate of return necessary to attract capital of different kinds of investment at any given time is the amount of risk involved.

Data on average earnings rates for enterprises in particular industries are available in financial services, trade association studies, and government publications. Care should be exercised in the use of such data to be sure that they are applied to comparable situations, for example, that the earnings rate consistently is assumed to be either before or after income taxes. We shall assume for purposes of illustration that a reasonable normal rate of return for Reed Company is 10% **after income taxes**.

Estimate of Future Superior Earnings

The amount of estimated future superior earnings may be defined as the amount of earnings expected in excess of normal earnings on the current fair value of identifiable tangible and intangible net assets.

All variables necessary to compute the estimated future superior earnings of Reed Company have been discussed and now can be illustrated. The current fair value of Reed Company's identifiable net assets is \$820,000 (see data on page 540), and its average future earnings are estimated at \$106,000. Because a 10% after-tax rate of return is sufficient to attract an investment in the company, estimated future superior earnings may be determined as illustrated on top of page 543. This computation shows that $\$820,000 \times 10\% = \$82,000$ a year is necessary to support a valuation of \$820,000 for the identifiable net assets of Reed Company. Because the company's prospects are for earnings in excess of \$82,000, the source of this excess earning power must be the unidentifiable intangible assets (goodwill) that enable the company to earn a higher-than-normal rate return.

Estimate of Present Value of Superior Earnings - The Final Step

A number of different methods may be used to value the estimated future superior earnings, and thus arrive at an estimate of goodwill. Four widely used methods are illustrated below:

Method 1 *Estimated future superior earnings are capitalized at the normal rate of return.*

One assumption is that the superior earnings of \$24,000 a year, as determined above, will continue unimpaired into the future and that this prospect is attributable entirely to the existing resources of Reed Company. The annual superior earnings are capitalized to answer the following question: How much capital should be invested if the annual return on the investment is \$24,000 in

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perpetuity, and the desired rate of return is 10% a year? Under this approach, goodwill is estimated at 240,000 as follows:

There are serious flaws in the assumptions on which this method rests. It may be reasonable to forecast that a business enterprise will earn a 10% return on its net assets over a very long period of time, but the assumption that superior earning power will persist in perpetuity in the face of competitive pressures and the hazards of free enterprise is optimistic, to say the least. Furthermore, even if superior earnings do continue, it seldom is possible to trace their origin to conditions that existed at the time of acquisition. The forces that erode superior earnings are such that a persistent ability to earn a higher-than-normal rate of return ultimately will be due to some additional propellant in the form of research, innovations, efficiency, and strategy on the part of the new owners and management.

Method 2 *Estimated future superior earnings are discounted for a limited number of years to determine the present value of such earnings.*

The estimate of goodwill may be modified in several ways to allow for the uncertain nature of superior earnings. One approach is to assume that any estimated future superior earnings will continue for a **limited period**, say, three five, or ten years. The **present value** of a given series of superior earnings at a given rate of return can be computed by the use of present value concepts described in Chapter 5. In Reed Company example, if estimated future superior earnings of \$24,000 a year will continue for a five-year period, the present value of this prospect on a 10% basis is approximately \$91,000, determined as follows:

Method 3 *Estimated future superior earnings are capitalized at a higher-than-normal rate of return.*

A variation of method 1 is to capitalize estimated future superior earnings at a higher discount rate than is used to capitalize normal earnings. For example, if the normal rate of return is considered to be 10%, a rate of, say, 20% or 30% may be used to capitalize superior earnings. The higher assumed rates of return allow for higher risk, because the prospect that superior earnings will continue unimpaired into the future is **much more uncertain** than the prospect of continued normal earnings. Referring once more to the Reed Company illustration, if superior earnings of 24,000 a year are capitalized at 30%, for example, goodwill is estimated at \$80,000 as follows:

Under this approach, the earnings prospects of the Reed Company have been divided into two layer - \$82,000 of normal earnings, and \$24,000 of superior earnings-and a different discount rate has been used to value each layer. Any number of different layers and any number of different discount rates might be used to estimate goodwill.

Method 4 *Estimated future superior earnings for a given number of years are purchased.*

Another approach to the estimate of goodwill is to multiply estimated future superior earnings by a number of years and to refer to the result as a "number of years of estimated future superior

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earnings purchased." For example, a goodwill estimate of \$120,000 may be described as "the purchase of five years of estimated future superior earnings of \$24,000 a year." Loose statements of this kind may obscure the real issues involved. As noted previously, the present value of five years of estimated future superior earnings of \$24,000 discounted at 10% is approximately \$91,000, not \$120,000. Therefore, no reason exists for paying \$120,000 for superior earnings of 120,000 to be received over a five-year period if money is worth 10% compounded annually.

Summary of methods

Uncertainty and subjectivity surround each of the variables involved in an estimate of goodwill. The probable amount of future earnings, the part that represents superior earnings, the length of time, and the appropriate rate of return to be used in the valuation of superior earnings-all are variables not subject to objective verification. They can be estimated only within a reasonable range of probability. The illustrated methods indicate the following possible range (from highest to lowest) in the estimated value of goodwill for Reed Company:

In a transaction involving the purchase of Reed Company, the value established for goodwill probably would be somewhere between \$240,000 and \$80,000, depending on the relative bargaining power of the purchaser and the seller. Inability to agree on a specific value for goodwill frequently results in an agreement to pay a minimum amount for goodwill, to be supplemented by additional payments **contingent** on future superior earnings of the acquired enterprise. Such agreements may raise numerous accounting questions. For example,

- (1) How should the future payments be recorded?
- (2) How should the earnings on which the contingent payments are based be measured?
- (3) How should future contingent payments be disclosed by the enterprise that may be required to pay them?

It sometimes is suggested that the market value of the shares of capital stock outstanding provides a basis for estimating the current fair value of a corporation. Thus, if Reed Company, whose net assets have a current fair value of \$820,000, has 200,000 shares of capital stock outstanding, quoted on the market at \$6 a share, this suggests that the company is worth \$1,200,000, and that goodwill is \$380,000 ($\$1,200,000 - \$820,000 = \$380,000$). This conclusion would have some merit if the market price of \$6 a share applied to the entire issue of 200,000 shares, or to a block representing a substantial and controlling interest in the company. However, only a small fraction of the total shares outstanding normally is offered for sale on the market at any given time. The market price of this small **floating supply** of capital stock can fluctuate widely, and are influenced by short-run factors that may be unrelated to the long-run prospects of the company. Furthermore, there is no quoted market price for the capital stock of the majority of small corporations. Stock price may be useful as evidence of **relative** values in the negotiation of business combination involving an exchange of capital stock, and they also may substantiate or cast doubt on estimates of goodwill reached independently. However, stock prices seldom are useful in the direct valuation of goodwill.

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Non-compete agreements

When a going business enterprise is purchased, the purchaser may pay an amount in excess of the current fair value of the identifiable net assets acquired. Typically, the excess is recorded as goodwill. There are situations, however, in which a part of the purchase price may be attributable to a restriction placed on the seller not to engage in a competing enterprise for a specified period of time. The purchaser of a retail store or a restaurant, for example, would not want the former owner to open a competing enterprise in the same geographic area.

A **non-compete agreement** is incorporated in the contract for the purchase of the business enterprise, as for example, "the seller agrees not to engage in the restaurant business in the City of Ames for a period of five years." Such an agreement obviously has value to the purchaser, and a reasonable portion of the purchase price should be assigned to it. Although the value of a noncompete agreement is difficult to determine, the purchaser and seller should be able to agree on a fair price. The value assigned to goodwill does not include the value assigned to the non-compete agreement, because the latter represents an **identifiable** intangible asset. For example, if a business enterprise with net assets of \$100,000 at current fair value is purchased for \$150,000, it appears that the purchaser paid \$50,000 for goodwill. If, however, the parties agree to place a value of \$30,000 on a non-compete agreement for five years, the purchase of the enterprise is recorded as follows:

The advantage of reducing the recorded value of goodwill is that the non-compete agreement is amortized at the rate of \$6,000 a year over its economic life of 5 years, and this amortization is a deductible expense for income tax purposes; amortization of goodwill is not a deductible expense in the computation of taxable income.

Controversy over amortization goodwill

Whether goodwill arising out of the purchase of a business enterprise should be amortized has been a controversial issue for many years. Even after issuance of **Opinion No. 17**, which required the amortization of goodwill acquired after October 31, 1970, many business executives and accountants have continued to question the amortization of purchased goodwill.

It has been argued that goodwill has an indefinite economic life and, therefore, should not be amortized until there is evidence that it no longer exists. Supporters of this view maintain that as long as earnings are sufficient to indicate the goodwill is unimpaired, it is a permanent asset. To amortize goodwill in the absence of a decline in earnings, it is argued, would obliterate the superior earnings which required the recording of the goodwill in the first place.

The argument against the amortization of purchased goodwill is particularly strong when earnings continue at a level which indicates that goodwill continues to exist. It is doubtful that continuing goodwill stems solely from conditions existing at the time of purchase. A more likely situation is that goodwill is maintained through the successful efforts of the new owners and management to stay ahead of competition. It is unlikely that the exact amount of original goodwill which has

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dissipated will be replaced by a new layer of internally developed goodwill. Retaining purchased goodwill intact in the accounting records would be an attempt to compensate for the accounting inconsistency of recording purchase goodwill and not recording internally developed goodwill. Expenditures for research and development and advertising necessary to maintain superior earning power are recorded as expenses. If purchased goodwill is amortized, there would be a duplication of expenses - the current expenditures incurred to build and maintain goodwill, and the periodic amortization of purchased goodwill.

The opposing view is that the amount paid for goodwill represents the purchase of a group of unidentifiable intangible assets and superior earnings for a limited number of years. It is argued that goodwill does not last forever and that the realization of superior earnings is not income to the new owners but rather a recovery of capital. Amortization of purchased goodwill is supported on practical grounds, because the value of goodwill is likely to become zero at some future date. Thus, the investment in goodwill should be accounted for on the same basis as any other productive asset having a limited economic life. If expectations were realized, that is, if earnings continued unchanged for the period of years used to estimate and amortize purchase goodwill, the result of amortization might be the reporting of less-than-normal earnings on the investment of the new owners during the amortization period. This squares with reality, because the payment for superior earnings make their ultimate emergence a **return of invested capital**, not income.

Both sides in this controversy agree that goodwill should be written down in the face of clear evidence that it is overstated. If superior earnings are eroded by competitive pressures and other economic conditions, the gradual disappearance of goodwill should be recognized as an expense.

Exercises on intangible assets.

1. In January 1997, Avon Company purchased for Kshs 8.5M a patent for a new consumer product. At the time of purchase, the legal life of the patent was 17 years. Because of the competitive nature of the product, the patent was estimated to have an economic life of 10 years. During 2001, the product was removed from the market under government order because of potential health hazard present in the product.

Compute the amount that Avon should debit to expense in year 2001, assuming that amortization is recorded at the end of each year.

2. Thika Limited incurred research and development costs in year 2000 as follows: Kshs

Material used in R&D projects	1,600,000
Equipment acquired which will be used in future R&D projects	8,000,000
Depreciation for year 2000 on above equipment	2,000,000
Labour costs of employees involved in R&D projects	4,000,000
Consulting fees paid to outsiders for R&D projects	400,000
Indirect costs reasonably allocable to R&D projects	800,000
Fully reimbursable R&D costs	775,000

Compute the amount of R&D costs that should be reported as an expense in the income statement for year 2000.

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3. From the following list of accounts, prepare the intangible assets section of Maziwa Limited's balance sheet:

	Kshs
Deposits with advertising agency that will be used to promote goodwill	550,000
Organisation costs	600,000
Discount on bonds payable	850,000
Excess of costs over net assets of purchased subsidiary	450,000
Patents	750,000
Franchise	950,000
Marketing costs of introducing new product	760,000
R&D costs expected to benefit future accounting periods	1,200,000

4. "If all the individual assets and liabilities of a business enterprise are identified and valued properly, goodwill will not exist." Do you agree with this quotation.
5. What expenditures are generally included in organisation costs?
6. Mombasa Limited applied for and received a patent on a manufacturing process. The legal fee and patent application fee totaled Kshs 500,000. Research expenditures leading to the patent were estimated at Kshs 1,200,000. Shortly after the patent was issued, the company spent Kshs 650,000 in legal fees against a suit in which it was claimed that Mombasa Limited' patent infringed on the patent rights of a competitor.
- (a) At what amount should the patent be recorded in the company's accounting records
 - (b) What is the legal life of this patent
 - (c) What factors should be considered in the determination of the patent's economic life?
1. Why is it more difficult to identify and determine the cost of intangible assets than that of tangible assets? What are some of the similarities between tangible and intangible assets?

Courtesy of Rodgers Wesonga

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CHAPTER NINE LONG-TERM INVESTMENTS

In Chapter 6 we discussed short-term investments, such as investments in shares of General Motors common stock or in American Telephone bonds. Such investments can be converted quickly to cash and are classified as current assets. Many business enterprises (termed **investors**) also make long-term investments in corporate securities to create close business ties with other companies (termed **investees**). These long-term investments are not current assets, because they do not represent resources available to meet working capital needs.

The basis of distinction between the asset categories of short-term investments (Chapter 6) and long-term investments lies in the nature and purpose of the investment. Investments that are readily marketable and which can be sold without disrupting business relationships or impairing the operating efficiency of the business enterprise are classified as current assets. Investments made to foster operational relationships with other enterprises are classified as long-term investments. Also, investments that do not meet the test of ready marketability are considered long-term, even if these investments do not promote business relationships. Long-term investments usually are listed below the current asset section of the balance sheet.

Objectives of long-term investments

A business enterprise may make long-term investments in the securities of other corporations for many reasons. For example, these investments may be used to create close ties to major suppliers or to retail outlets. The rights of ownership inherent in common stock investments give an investor in such securities a degree of influence or control over the management of the investee company. Thus, many enterprises use investments in common stock as a company with a strong cash position, or diversifying by acquiring ownership interest in investees in other industries.

Consolidated financial statements

A Company that acquires a controlling interest in the common stock of another company is termed the **parent company**, and the controlled company the **subsidiary**. The investment in the common stock of the subsidiary is a long-term investment for the parent company. In addition to the separate financial statements prepared by the parent company and by the subsidiary, **consolidated**

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financial statements also are prepared. Consolidated finance statements ignore the legal concept that subsidiaries as single economic entity.

Viewing both enterprises as a single economic entity is an alternative to treating the subsidiary as an investment owned by the parent company. The circumstances in which consolidated financial statements are appropriate and the manner in which consolidated financial statements are appropriate and the manner in which they are prepared are topics disclosed in **Modern Advanced Accounting** of this series.

Cost at acquisition

The cost of an investment in securities includes the purchase price plus brokerage fees, transfer taxes, and any other expenditures incurred in the transaction. If assets other than cash are given in payment for the securities and the current fair value of such noncash assets is unknown, the current market price of the securities may be used to establish the cost of the securities and the value of the noncash assets given in exchange. When neither a market price for the securities nor current fair value of the assets given in exchange is known, accountants must rely on independent appraisals to establish values for recording the transaction.

If two or more securities are acquired for a lump sum, the total cost should be allocated among the various securities. If the various securities purchased are publicly traded, the existing market prices serve as the basis for apportionment of the total cost. This type of cost apportionment is termed **relative sales value allocation**.

Assume, for example, that X Company acquire from Y Company 100 units, of five common shares and one preferred share each, at a price of \$240 a unit, when the common stock is selling for \$30 and the preferred stock for \$100 a share. The portion of the cost allocated to the common stock is $\$24,000 \times \frac{150}{250}$, or \$14,400, and the portion allocated to the preferred stock is

$\$24,000 \times \frac{100}{250}$, or \$9,600. If only one class of the stock is publicly traded, that class usually will be recorded at its market price, and the remaining portion of the cost is considered the cost of the other class. When neither class of stock has an established market, the apportionment of the cost may have to be delayed until current fair values of the securities can be established.

Accounting for investment in stocks

¹ Cost method. Investment income consists only of dividends received.

² Equity method. Investment income consists of the investor's proportionate share of the investee's net income.

³ Market value method. Investment income includes dividends received and changes in the market value of the investment.

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Measuring return on investment

What is the “return” on an investment in common stock? One point of view is that the investor’s return consists of the stream of dividends received from the investment. A second point of view is that the investor’s return consists of a proportionate share of the net income (minus preferred dividends, if any) of the investee, with regard to whether this income is distributed during the accounting period in the form of dividends. Supporting this second viewpoint is the fact that the earnings investee, with a resultant increase in stockholders’ equity. A third interpretation of the investor’s return consists of the dividend received plus (or minus) the change in the market value of investment.

Three different accounting methods exist, depending on which return an investor wishes to measure. These methods are:

The market value method (as an alternative to the cost method) is illustrated for short-term investments in Chapter 6. However, the market value method is much less appropriate for long-term investments. By definition, long-term investments are not held to take advantage of short-term fluctuations in market prices. When an investor intends to hold an investment in securities for long periods of time, the daily changes in market price lose significance. Therefore, the cost and equity methods generally are used to account for long-term investments in common stock. **Accounting for dividends received**

When an investor owns only a small portion (for example, less than 20%) of the total outstanding common stock of an investee, the investor has little or no control over the investee. In this case, the investor cannot influence the investee’s dividend policy, and the only portion of the investee’s income which reaches the investor is the dividends declared and paid by the investee. Thus, when the investor has little or no control over the investee, the dividends received represent the only return realized by the investor. Under these circumstances, the cost method of accounting for the investment is appropriate.

The payment of dividends on common stock is a discretionary act, requiring that the board of directors first declare the dividend. For this reason, investors should not accrue dividend. For this time as they do interest revenue on a bond. There are three acceptable alternatives for timing the recognition of dividend revenue: (1) when the dividend is declared (declaration date), (2) when the dividend “accrues” to the current stockholder even if the stock is subsequently sold (ex-dividend date), or (3) when the dividend is received (payment date). For the purpose of consistency, all illustrations in this chapter recognize dividend revenue on the date the dividend is received.

Not all dividends received represent revenue to the investor. Some times corporations may pay dividends in excess of net income. In such cases the amount by which the cash distribution exceeds total earnings to date is considered a return of capital, termed a liquidating dividend, revenue.

Some accountants have suggested that, from the viewpoint of any given stockholder, a liquidating dividends may be deemed to have occurred if dividends received exceed total net income earned

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subsequent to the date the investment was acquired. Practical application of such a concept would be difficult, because corporations do not measure net income on a daily basis, whereas the acquisitions of shares of common stock by individual investors occurs throughout the year. Moreover, some large investors make a series of acquisitions of an investee's common stock without disrupting the market price of the stock. Only in very special circumstances would an investor be able to determine that a dividend received represented net income earned prior to the date of a specific acquisition of common stock.

For income tax purposes, liquidating dividends are defined with respect to the investee paying the dividend, rather than with respect to individual investors. Tax laws recognize liquidating dividends only to the extent that total dividends paid exceed total net income **over the life of the investee** paying the dividend.

Applying the cost method of accounting

When the cost method of accounting is used, the investment account is maintained in terms of the cost of the common stock acquired. Revenue is recognized only to the extent of dividends received, which do not exceed the cumulative earnings from the date the common stock was acquired. Changes in the net investee are ignored unless a significant and permanent impairment of value of the investment occurs. Finally, long-term investments in marketable equity securities may be written down to a lower of cost or market as required by **FASB Statement No. 12**. The three events which may cause a departure from the cost basis are discussed below. **Liquidating Dividends**

When the cost method is used, ordinary dividends received from an investee are recorded the same as dividends on any corded by the investor only when and to the extent that dividends are distributed or when realized as a gain or loss at the time the stock is sold. However, any liquidating dividends received are recorded by credits to the investment account.

To illustrate the accounting for a liquidating dividend, assume that Dunn Company acquired 15% of the common stock of Blue Company early in Year 1. During Year 1, Blue Company reported net income of \$100,000 and declared and paid a cash dividend of \$150,000. Because the dividend exceeded the net income of Blue Company for the period Dunn owned Blue's common stock, Dunn should the dividend as follows:

Permanent Decline in Value of Investment operating losses of the investee that reduce the investee's net assets substantially and seriously impair its future prospects are recorded as losses by the investor. A portion of the long-term investment has been lost and this fact is recorded by reducing the carrying amount of the investment. The following excerpt from **FASB Statement No. 12** supports this approach:¹

If the decline is judged to be other than temporary, the cost basis of the individual security shall be written down to a new cost basis and the amount of the write-down shall be accounted for as a realized loss. The new cost basis shall not be changed for subsequent recoveries in market value.

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The journal entry to record such a loss is:

The realized loss is included in the computation of income before extraordinary items in the income statement.

¹The length of time the security has been owned

²The length of time market value of the security has been below cost and the extent of the decline

³The financial condition and prospects of the investee

⁴The financial condition of the investor

⁵The materiality of the decline in value of the investment in relation to the net income and stockholders' equity of investor

Valuation at Lower of Cost or Market

Special accounting procedures are required when the aggregate market value of a noncurrent portfolio of marketable equity securities. Accounted for by the **cost method**, is below cost. Such a portfolio should be valued at the lower of cost or market. However, in contrast to the valuation of the current portfolio of marketable equity securities discussed in Chapter 6, the unrealize loss in value of a noncurrent portfolio of marketable equity securities is not **included in net income**.

Instead the unrealized loss is reported as a **reduction of stockholders' equity**.²

To illustrate the accounting for long-term investments in marketable equity securities at lower cost or market, assume the following: Early in Year 1, Investor Company made long-term investments of \$200,000 in the common stock of publicly owned corporations. The market value of the investments was \$160,000 at the end of Year 1 and \$184,000 at the end of Year 2. The journal entries to value the long-term investments in marketable equity securities at lower of cost or market at the end of Year 1 and at the end of Year 2 are:

When a long-term investment in marketable equity securities is sold at a price below original cost, a **realized loss** is recorded; however, no journal entry is made in the unrealized loss and valuation accounts until the end of the accounting period. For example, assume that on July 10, Year 3, Investor Company sold long-term investments which cost \$100,000 for \$75,000, and that at the end of Year 3 the market value of the investments on hand is \$90,000. The journal entry for the sale and the adjusting entry at the end of Year 3 are illustrated below:

The balance sheets of Investor Company at the end of each year include the following information:

Applying the equity method of accounting

When the investor owns enough common stock of the investee to exercise significant influence over the investee's management, the dividends paid by the investee no longer may be a good measure of the return on the investment. This is because the investor may influence the investee's dividend policy. In such a case, dividends paid by the investee may reflect the investor's income tax considerations and cash needs, rather than the profitability of the investment.

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For example, assume that an investor owns 100% of the common stock of an investee. For two years the investee is very profitable but pays no dividends, because the investor has no need for additional cash. In the third year, the investee has a net loss but pays a large cash dividend. It would be misleading for the investor to report no investment income while the investee was operating profitably, and then to show large investment income in a year when the investee operated at a loss.

The investee need not be fully owned for the investor to have a significant degree of influence. When the common stock of the investee is widely held, an investor with much less than 50% of the common stock may have effective influence over the investee, because it is doubtful that the remaining outstanding shares will vote as an organized block.

When the investor has a significant degree of influence over the investee, the equity method of accounting more fairly presents the benefits accruing to the investor than does the cost method. When the investor has little or no influence over the investee, the benefits received by the investor may be limited to the dividends received, indicating the cost method of accounting to be more appropriate. The key criterion in selecting between the methods is the degree of influence the investor is able to exercise over the investee.

To achieve a degree of uniformity in accounting practice, the Accounting Principles Board took the position in Opinion No. 18 that “an investment (direct or indirect) of 20% or more of the voting stock of an investee should lead to a presumption that in absence of evidence to the contrary an investor has the ability to exercise significant influence over an investee.”³ Thus, investments representing 20% or more of the voting stock usually are accounted for by the equity method of accounting. However, if the investor owns 20% or more of the voting stock but is unable to exercise significant influence over less than 20% usually are accounted for by the cost method of accounting, unless clear-cut ability to influence the operating and financial policies of the investee can be demonstrated. Investments in preferred stock should be accounted for by the cost method, because preferred stockholders usually do not have either voting rights or a residual equity in net income.

When the equity method of accounting is used, an investment in common stock initially is recorded at the cost of the stock acquired, but is adjusted for changes in the net assets of the investee subsequent to acquisition. The investor’s proportionate share of the investee’s net income is recorded as investment income, causing an increase in the investment account. If the investee’s net income includes as extraordinary (if material in amount to the investor), rather than as ordinary investment income. Dividends paid by the investee are recorded by the investor as a conversion of the investment to cash, causing the investment account to decrease.

Illustration of Equity Method of Accounting

To illustrate the equity method of accounting, assume that Investor Company purchased 40% of the common stock of Lee Company for \$300,000, which corresponded to the underlying carrying amount of Lee’s net assets. During the subsequent accounting period, Lee reported net income of

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\$70,000 (including a \$10,000 extraordinary gain) and paid dividends of \$30,000. Investor Company accounts for its investment in Lee Company as follows:

Note that the net effect of Investor's accounting for Lee's net income and dividends was to increase the investment account by \$16,000. This corresponds to 40% of the increase in Lee's net assets during the period $[(\$70,000 - \$30,000) \times 40\% = \$16,000]$.

Special Problems in the Application of Equity Method

Two problems often arise in the application of the equity method of accounting. First, intercompany profits and losses resulting from transactions between the investor and the investee must be eliminated until realized by a transaction with an unaffiliated entity. This special problem is discussed in Modern Advanced Accounting of this series. Second, when the acquisition cost of an investment differs from the carrying amount of the underlying net assets, adjustments may have to be made to the investment income recorded by the investor.

Cost in Excess of Equity Acquired

Often an investor will pay more than the underlying equity of an investment because current fair values of the investee's assets may be larger than their carrying amounts, or because the investee has unrecorded goodwill. In either case, this excess of cost over the underlying equity will benefit the investor only over the economic lives of the undervalued (or unrecorded) assets.

To the extent that the excess cost was incurred because of implied goodwill, the amount should be amortized over the estimated economic life of the goodwill. The Accounting Principles Board took the position that amounts paid for goodwill should be amortized over a period of not more than 40 years.⁴ If the excess of the cost over the underlying equity is small, it usually is amortized as goodwill, rather than allocated to specific assets.

Cost Less than Equity Acquired

In some cases, an investor may acquire an investment in common stock at a cost less than the underlying equity. In this event, it should be assumed that specific assets of the investee are overvalued. If these assets have limited economic lives, the investor allocates the excess of the underlying equity over cost to investment income over the economic lives of the assets. The journal entry to record this amortization is given on top of page 577.

Note that this adjustment increases investment income. The rationale for this action is that the investee's reported net income is understated, because the investee has recorded depreciation or amortization based on overstated asset values.

Summary of procedures under the Equity Method of Accounting

Accounting procedures under the equity method may be summarized as follows:

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1. The investment initially is recorded at cost.
2. That investor subsequently records its proportionate share of the investee's net income (after elimination of intercompany profits) by a debit to the investment account and a credit to Investment Income. In event of a loss, Investment Loss is debited and the investment account is credited.
3. The investor views its share of dividends paid by the investee as a conversion of the investment to cash. Thus, the investor debits Cash and credits the investment account.
4. The investor adjusts the recorded amount of investment income or loss by the amortization of any excess of cost over the underlying equity associated with depreciable assets or goodwill. This adjustment consists of a debit to Investment Income (or Loss) and a credit to the investment account.
5. The investor adjusts the recorded amount of investment income or loss by the amortization of any excess of the underlying equity over the cost by a debit to the investment account and a credit to Investment Income (or Loss).

Comparative illustration of the cost and equity methods of accounting

To illustrate the differences in the cost and equity methods, assume that on January 2 of the current year Investor Company acquired 4,000 shares (20%) of the common stock of Investee Company for \$1,000,000. At the date of acquisition, the carrying amount of Investee Company's net assets was \$4,550,000. Investor was willing to pay more than the underlying equity for the investment because it was willing to pay more than the underlying equity for the investment because it was estimated that Investee owned land worth \$100,000 more than its carrying amount, and enough goodwill to make a 20% interest in Investee Company worth the \$1,000,000 cost.

The excess of the cost of the investment over the underlying equity is analyzed on top of page 578.

The undervalued depreciable plant assets have an average remaining economic life of 10 years, and Investor Company's policy with respect to goodwill is to amortize it over 40 years.

During the current year, Investee Company reported net income of \$430,000, after an extraordinary loss of \$50,000, and declared and paid dividends of \$200,000 at year-end. Investor's accounting for its investment in Investee during the year is illustrated on page 579 under the cost and equity methods of accounting.

Note that no adjustment is made under either the cost or the equity method for the \$20,000 excess of cost over the underlying equity representing Investor's 20% interest in Investee's undervalued land. This is because land is not depreciated. The results for the current year are illustrated below for both methods of accounting:

Accounting for investments in bonds

A bond contract represents a promise to pay a sum of money at maturity and a series of interest payments during the term of the contract. Investors acquire corporate bonds to earn a return on investment. The effective rate of return (yield) on bonds to investors is determined by the price

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investors pay for the securities (because the terms of the contract are fixed). The yield on the bonds to investors may differ from the effective interest cost to the borrower because the bonds may have been issued at an earlier date at a different price.

Computation of purchase price of bonds

The cost of an investment in bonds is the present value for the future money receipts promised in the bond contract, measured in terms of the market rate of interest prevailing at the time of investment. The stated (coupon) rate of interest in the bond contract measures the cash to be received semiannually by the investor. If the rate of return demanded by investors is exactly equal to the coupon rate, the bond can be acquired at the face amount. If the market rate of interest exceeds the coupon rate, the bond can be acquired at a discount, because the investor is demanding a higher return than the bond contract offers; therefore, to equate the yield on the bond with the market rate of interest, the bond is acquired at a price below face amount. If the market rate of interest is below the coupon rate, the investor will be willing to pay a premium for the bond, that is, a price above face amount.

To illustrate the computation of the purchase price of bonds, assume that \$200,000 of 7% bonds maturing in 15 years are purchased by Kane Company to yield 8% compounded semiannually. The bonds pay interest semiannually starting six months from date of purchase. Because the market rate of interest exceeds the coupon rate, the bonds are purchased at a discount, as shown below (using the Appendix at the end of the book):

If the market rate of interest was only 6% compounded semiannually, the bonds paying semiannual interest at 7% a year would be purchased at a premium, as shown below:

Acquisition between interest dates

Interest on a bond accrues with the passage of time in accordance with the provisions of the contract. The issuing corporation pays the contractual rate of interest on the stated day to the investor owning the bond on that day. The investor who acquires a bond between interest dates must pay the owner the market price of the bond plus the interest accrued since the last interest payment. The investor is paying the owner of the bond the interest applicable to the first portion of the interest period and will in turn collect that portion plus the additional interest earned by holding the bond to the next interest payment date.

Illustration

On July 1, an investor acquired 10 bonds of Ray Company, which had been issued several years ago. The bond contract provides for interest at 8% a year, payable semiannually on April 1 and October 1. The market rate of interest is higher than 8% at the present time, and the bonds are

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currently quoted at ⁹⁷/₄ plus accrued interest for three months. The journal entry for the investor to record the acquisition of the 10 bonds is:

Discount and premium on Investments in bonds

On the date of acquisition of bonds, the investment account usually is debited for the cost of acquiring the bonds, including brokerage and other fees, but excluding the accrued interest. The use of a separate discount or premium account as a valuation account is acceptable procedure; however, it seldom is used. The subsequent treatment of the investment might conceivably be handled in one of three ways:

- (1) The investment might be carried at cost, ignoring the accumulation of discount or amortization of premium;
- (2) The investment account might be revalued periodically to reflect market value changes; or
- (3) The discount or premium might be accumulated or amortized to reflect the change in the carrying amount of the bonds based on the effective rate of interest prevailing at the time of acquisition.

The first alternative (the cost basis) is used primarily in accounting for short-term bond investments, as discussed in Chapter 6, for convertible bonds, and for other bonds for which the discount or premium is insignificant. The discount or premium on convertible bonds seldom is related to the level of interest rates, but rather reflects the effect of the price of the security to which the bond is convertible. These securities are subject to wide price movements related to changes in the market price of common stocks; therefore, the amortization of premium or accumulation of discount does not seem appropriate.

The second alternative (valuation at market) is not in accord with the present interpretation of the realization principle or the concept of conservatism, especially during periods of rising bond prices. Changes in market prices of bonds held as long-term investments may be of less significance to the investor than changes in prices of short-term investments, because the longterm investments frequently are held to maturity, at which time market price and face amount of the bonds are equal. When the investment in bonds is in jeopardy because of serious cash shortages of the issuing corporation, it generally is acceptable to write the investment down to its expected net realizable value and to recognize a loss.

The third alternative (the systematic accumulation and amortization) is the preferred treatment for long-term investments in bonds. This approach recognizes that the interest revenue represented by the discount, or the reduction in interest revenue represented by the premium, does not come into being at maturity of the bonds, but accrues over the term of the bonds. The interest revenue should be consistent with the yield on the bonds at the date of acquisition. This method is consistent with the principle that requires assets to be recorded at cost.

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Interest revenue

The periodic interest payments provided for in a bond contract represent the total interest revenue to an investor holding a bond to maturity only if the investor acquired the bond at its face amount. If an investor acquires a bond at a premium, the amount received on maturity of the bond will be less than the amount of the initial investment, thus reducing the cumulative interest revenue by the amount of the received at maturity will be larger than the initial investment, thereby increasing the cumulative interest revenue by the amount of the discount.

When an investor intends to hold bonds to maturity, there is little logic in treating the discount or premium as a gain or loss occurring on the maturity date. Rather, the increase in the carrying amount of the bonds as a discount disappears should be viewed as part of the revenue accruing to the investor over the entire period the bonds are owned. Similarly, the decrease in value when a premium disappears is a cost the investor is willing to incur over the entire holding period to receive periodic interest payments higher than the market rate prevailing when the bonds were acquired. Thus, the amount of the discount or premium is viewed as an integral part of the periodic interest revenue earned by the investor. The accumulation of a discount increases periodic interest revenue, and the amortization of a premium decreases periodic interest revenue.

An extreme illustration of this concept has occurred for certain government savings bonds which provided no periodic interest payments at all. Instead, these bonds were issued at a large discount, and the gradual growth in the redemption value of the bonds toward their maturity value (accumulation of the discount) was the investor's only return. Although the investor received no cash proceeds until the bonds matured, interest revenue was being earned. To measure the periodic interest revenue, the accumulation of the discount had to be recognized as interest revenue over the term of the bonds.

Methods of discount accumulation or premium amortization

The methods of amortization for bond discount and bond premium by the issuer are discussed in Chapter 15. These methods present precisely the same problem for the investor as for the issuer. The purpose of accumulating the discount or amortizing the premium is to reflect accurately the interest revenue derived from the investment in bonds.

Interest method

The interest method produces a constant rate of return on the investment in bonds. That is, the periodic interest revenue always represents the same percentage return on the carrying amount of the investment. Thus, when a discount is being accumulated and the investment account is increasing, the interest revenue recognized each interest period also increases. This is accomplished by accumulating an ever-increasing portion of the discount each period. To apply the method, the interest revenue is computed for each interest period by multiplying the balance of the investment account by the effective interest rate at the time the investment was made. The accumulation of the discount (or amortization of the premium) thus is the difference between the

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periodic cash receipt and the interest revenue for the period computed by the effective rate of interest.

Straight-Line Method

Under the straight-line method, the discount or premium is spread uniformly over the term of the bonds. Although the bonds may be sold by the investor or redeemed by the issuer prior to maturity, the accumulation or amortization always is based on the years remaining to maturity. The straight-line method is simple to apply and avoids the necessity for determining the yield rate. The primary objection to the straight-line method is that it produces a constant amount of interest revenue each accounting period, which results in an uneven rate of return on the investment.

Illustration

The computation of periodic discount to be accumulated or premium to be amortized and the related journal entries will be illustrated with the examples for Kane Company given on page 580. The Kane Company examples involved

- (1) the acquisition of \$200,000 face amount of 7% bonds maturing in 15 years (or 30 semiannual periods) to yield 8% compounded semiannually, and
- (2) the acquisition of the same bonds to yield 6% compounded semiannually. The journal entries to record the investment, receipt of interest for the first year, and the related accumulation or amortization under the interest method and the straight-line method are presented below and on page 585. (All computations are rounded to the nearest dollar.)

When the bonds are acquired at a discount and the interest method of accumulation is used, the investment account is increased to \$183,016 at the end of the first six-month period; therefore, interest revenue for the second six-month period is \$7,321 ($\$183,016 \times 4\% = \$7,321$), which required \$321 of the discount to be accumulated.

When the bonds are acquired at a premium and the interest method of amortization is used, the investment account is reduced to \$219,188 at the end of the first six-month period; therefore, interest revenue for the second six-month period is \$6,576 ($\$219,188 \times 3\% = \$6,576$), which required \$424 of the premium to be amortized.

Interest revenue on bond investments, just as interest revenue on any other investment, is accrued only at significant dates. The significant dates are:

- (1) interest payment dates,
- (2) the end of the investor's accounting period, and
- (3) the time of any transaction involving the particular investment which does not coincide with a regular interest payment date. The interest revenue on bond investments must be accrued, therefore, at the end of the accounting period and before the bonds are sold. The discount also

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should be accumulated or the premium amortized in accordance with whatever method of accumulation or amortization is used.

Special problems in accounting for investments in securities

Cost identification

Investments in securities may pose a problem as to which costs should be offset against revenue in the period of sale. For example, assume that an investor acquires 1,000 shares of Z Company common stock at a price of \$80 a share, and 1,000 shares at \$90 a share. Several years later, the investor sells 1,000 shares of Z Company common stock at a price of \$80 a share, and 1,000 shares at \$90 a share. Several years later, the investor sells 1,000 shares of Z Company common stock for \$84 a share. Should the investor recognize a \$4,000 gain or a \$6,000 loss?

The answer to this question requires making a cost flow assumption, as with inventories. Because securities usually are identified by a certificate number, it would be possible to use specific identification of stock certificates to establish the cost of the 1,000 shares sold. However, an alternative cost flow include:

- (1) fifo - the last shares acquired are assumed to be the first ones sold; and (2)
- lifo - the last shares acquired are assumed to be the first ones sold; and
- (3) weighted-average cost - each share is assigned the same cost basis.

Income tax rules require the use of either the specific identification method or the fifo method to measure taxable gain or loss. Neither lifo nor weighted-average cost is an acceptable method for income tax purposes. The specific identification method usually is more advantageous for income tax purposes, because it allows the investor to select for sale those securities which will lead to the most desirable tax consequences. For financial accounting purposes, most investors use the same method of cost selection used for income tax purposes, to simplify record keeping. From a theoretical viewpoint, however, weighted average is the only cost flow assumption that recognizes the economic equivalence of identical securities. In our illustration of successive purchases of the common stock of Z Company at different prices, it is undeniable that each share of Z Company common stock owned has exactly the same economic value regardless of the price paid to acquire it. The weighted-average cost flow assumption recognizes the economic reality that, except for income tax purposes, it makes no difference which 1,000-share certificate is sold and which is retained.

Accounting for stock dividends and stock splits

Stock dividends and stock splits do not result in revenue to investors. The income tax regulations are in agreement with financial accounting procedures on this point.⁵

Since a shareholder's interest in the corporation remains unchanged by a stock dividend or splitup except as to the number of share units constituting such interest, the cost of the shares previously

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held should be allocated equitably to the total shares held after receipt of the stock dividend or split-up. When any shares are later disposed of, a gain or loss should be determined on the basis of the adjusted cost per share.

The accounting procedure of the investor to record receipt of additional shares from a stock split usually is confined to a memorandum entry which indicates the number of shares of stock received and the new cost per share.

Property dividends

When a corporation distributes a dividend in the form of merchandise, securities of other corporations, or other noncash assets, the investor records the property received at its current fair value. Income tax regulations also require the use of current fair value for property dividend received.

Stock purchase warrants and stock rights

A stock warrant is a certificate issued by a corporation conveying to the holder rights to purchase shares of its stock at a specified price within a specified time period. A single right attaches to each share of outstanding stock, and several rights usually are required to purchase one new share at the stipulated price. For example, when rights are issued, the owner of 100 shares of common stock will receive a warrant representing 100 rights and specifying the number of rights required to purchase one new share of stock. The term of these rights usually is limited to a few weeks. The rights must be exercised or sold before the expiration date or they become worthless.

Accounting for Stock Warrants Acquired by Purchase

The accounting problems involved when an investor buys warrants are similar to those relating to the acquisition of any security. The purchase price, plus brokerage fees and other acquisition costs, is debited to Investment in Warrants, and the credit is to Cash. When warrants are acquired as a part of a package, the total cost must be allocated to the various securities included in the package, based on relative market values.

When the warrants are used to acquire stock, the initial cost of the warrants used plus the cash paid is the cost of the stock. The Investment in Stock account is debited; Cash and Investment in Warrants are credited. If the market price of the stock differs from this combined cost, this fact is ignored until the stock is sold, at which time a gain or loss is recorded.

Accounting for Stock Rights

Stock rights are distributed to the stockholders of a corporation in proportion to their holdings. The receipt of stock rights can be compared to the receipt of a stock dividend. The corporation has not distributed any assets; instead, the way has been opened for an additional investment by

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the present stockholders. Until the stockholders elect to exercise or sell their rights, their investment in the corporation is represented by

- (1) shares of stock that have been purchased, and
- (2) the right to acquire additional shares of stock at a price below the current market price. The cost of the original investment consists of the cost of the shares and the rights; therefore, the cost of the original investment is apportioned between these two parts of the investment on the basis of relative market values. The stock will trade in the market on a “rights-on” basis until the ex-rights date, at which time the stock sells “ex-rights,” and the warrants for the rights have a market of their own. Relative market value allocation may be used to apportion the cost between the shares of stock and the stock rights as follows:

Convertible securities

An investor may invest in bonds or preferred stocks that are convertible to the common stock of the investee at the option of the investor. The characteristics of convertible securities are discussed in Chapter 15 and 17. At this point, we shall consider the action to be taken by investors who exercise the conversion option and receive common stock in exchange for convertible bonds or convertible bonds or convertible preferred stock.

The market value of the common stock received may differ materially from the carrying amount of the converted securities. However, it is virtually universal practice to assign the carrying amount of the convertible security to the common stock acquired in exchange. Thus, no gain or loss is recognized at the time of conversion. This treatment is supported by the theoretical argument that investors contemplate conversion when they acquire a convertible security. Thus, no gain or loss is recognized until the common stock acquired by conversion is sold.

The following journal entry illustrates the conversion of an investment in Quincy Company bonds with a carrying amount of \$96,720 to Quincy’s common stock with a current market value of \$120,000.

Other long-term investments

Investments in special-purpose funds

Occasionally, a corporation accumulates a fund of cash, usually invested temporarily in securities, for a special purpose. The creation of the fund may be by voluntary action on the part of the management, or it may be required by contract. Funds generally are created to pay a liability or to acquire specific assets. In general, funds are treated as long-term investments only when they are established by contract, and the money invested is not available to management for general operating needs. A fund is classified as a current asset if it is created voluntarily and can be used for operating purposes.

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Accounting for funds

The transactions which must be accounted for in connection with fund accumulation and administration are:

- (1) The fund may be established and operated internally; or
- (2) The assets may be deposited with a trustee who is charged with receiving the deposit, investing the assets, collecting the revenue, paying the expenses, and accounting to the responsible officials for cash receipts and payments.

Typically, the funds which are created voluntarily are operated internally, whereas those created by contract are handled by a trustee. The periodic deposit to the fund generally is set in advance. It may be related to the level of operations, or it may be set either as a stated amount each period or as a stated amount less earnings on fund assets for the period. The method of determining the amount and time for the deposit generally can be found by referring to the document authorizing the establishment of the fund. In cases when the fund is committed irrevocably for the purpose designated, and cash actually is deposited with a trustee, the fund itself may not appear among the assets of the business enterprise, and the liability which is to be paid from fund assets may be excluded from the liabilities. This procedure is used most often when the liability does not exceed the fund balance, which means that the enterprise has no liability other than that for the periodic deposits stipulated in the contract. Most employee pension and benefit plans are of this type.

Bond sinking funds usually are included under long-term investments, and bonds outstanding are shown as a long-term liability. The sinking fund should not be offset against the bond liability. A sinking fund and other similar funds usually are included in the balance sheet as an asset even though trustees hold them.

One of the most common methods of accumulating a sinking fund is to deposit fixed amounts at periodic intervals. The periodic deposit is computed by use of an amount of annuity formula described in Chapter 5.

The transactions relating to the purchase and sale of securities, and the accrual and collection of revenue for the sinking fund, are accounted for in the same manner in which transactions in the general investments account are recorded,

Cash surrender value of life insurance policies

When a business enterprise is dependent on certain officers for direction and management, life insurance policies may be purchased on the lives of these officers with the enterprise named as the beneficiary. Certain types of insurance policies combine a savings portion of the insurance premium is reported in the balance sheet as an investment.

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The savings part of a life insurance policy is referred to as the cash surrender value of the policy. This is the amount the enterprise would receive in the event that the policies were canceled; this same amount also can be used as collateral for a loan.

The following data represent the first four years' experience of White Company, which has a \$100,000 life insurance policy on one of its officers:

From these limited data, we can readily see the increase in the asset and the decreasing annual cost of life insurance. The journal entries for the first two years are as follows:

In the event of death of the insured officer, White Company would collect the face amount of the insurance policy. The journal entry to record this event, assuming death occurred early in the third year, would be as follows:

For financial accounting purposes, the gain is included in income before extraordinary items in the income statement. For income tax purposes, the premiums paid on life insurance policies in which the business enterprise is the beneficiary are not deductible. Similarly, the gain on the settlement upon the death of the insured party is not taxable income.

Presentation in financial statements

Long-term investments that cannot be sold without impairing business relationships are classified as noncurrent assets, immediately following current assets.

Dividends and interest revenue normally are listed under the caption Other Revenue and are included in the determination of income before extraordinary items. When the equity method of accounting is used, ordinary investment income (or loss) also is included in Other Revenue, but the investor's share of any material extraordinary item of the investee retains its extraordinary nature and is classified as an extraordinary item by the investor. Because of the nature of long-term investments, gains and losses from sales occur relatively infrequently. A business enterprise with numerous long-term investments can expect occasional gains and losses from sales of these investments and generally should include such gains and losses in income before extraordinary items, under the caption of Other Revenue.

Exercises

1. Distinguish between the cost and equity methods of accounting for a long term investment in common stock. When is each appropriate?
2. Identify three events that necessitate a write down of a long term investment under the cost method of accounting.
3. Distinguish between the interest and the straight line methods of accumulating a discount and amortizing a premium on an investment in bonds.
4. Why is the cash surrender value of an insurance policy on the life of an official of a business enterprise carried as a long term investment in the balance sheet of the enterprise?

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